

burn_gekko: Stable Output Heads, Measured Limitations

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Local technical draft / experiment head-stability15

Abstract

One frozen multi-view foundation with newly trained camera-calibration and RGB reconstruction heads. An 800-update CPU study uses 32 training rooms and 8 disjoint development rooms. RGB completion reaches 21.09 dB hidden-pixel PSNR. Both heads pass bounded numerical stability and exact checkpoint replay; camera calibration still overfits and is not reliable. All figures use this foundation and one final set of attached head weights. This report presents a single selected training experiment in Burn, with fixed-teacher latent completion, co-visibility evaluation and declared correspondence readouts. Every table and visual binds to one checkpoint. Latent-space visualization is distinct from RGB synthesis; state-of-the-art performance is not established.

1 Model and task

An audited MIT V-JEPA 2.1 Base image encoder supplies final and block-6 features to six fusion blocks of width 384 with six attention heads. This phase starts from the audited own Pilot 12 checkpoint, resets both AdamW optimizers, and trains all 151 encoder tensors. Images are 256 by 256, batch size is 16, two references are provided and 90% of target patches are masked. The original fixed V-JEPA supplies latent targets. A fixed own Pilot 11 encoder supplies training-only final-feature preservation with weight 4.0; it is absent from inference. The latent, RI and image-homography objectives are retained. Seed is 829, trunk learning rate is $2e-5$, encoder rate is $4e-7$, warmup is 32 and the cosine horizon is 384 updates. The spatial residual is bounded by 25% of the centered block-6 norm. Matching uses a fixed 3 by 3 local probability centroid, plus the retained hard readout. An additional renderer-supervised bidirectional correspondence NLL has weight 0.1 and temperature 0.07. It pairs the dense target with a rotating reference using known visible patch-center projections. Source depth discontinuities, unknown depth, occlusion, out-of-frame and descriptor-border locations are excluded. Geometry supplies detached training labels only. All inference inputs are RGB. Known intrinsics enter the calibrated CPU pose solver after inference. No noncommercial pretrained weights or teachers are used. Attached RGB head: predicted 768-dimensional completion latent, 64-wide GELU MLP, sigmoid 16x16 RGB patches. Attached camera head: both directions of dense RGB-pair fusion, 4x4 spatial pooling per direction, shared 16-channel projection and 64-wide MLP; six rotation coordinates, signed translation direction and log normalized focal lengths. Camera labels and target RGB are loss/scoring targets only. Both heads are trained from scratch with separate AdamW optimizers; the foundation is frozen.

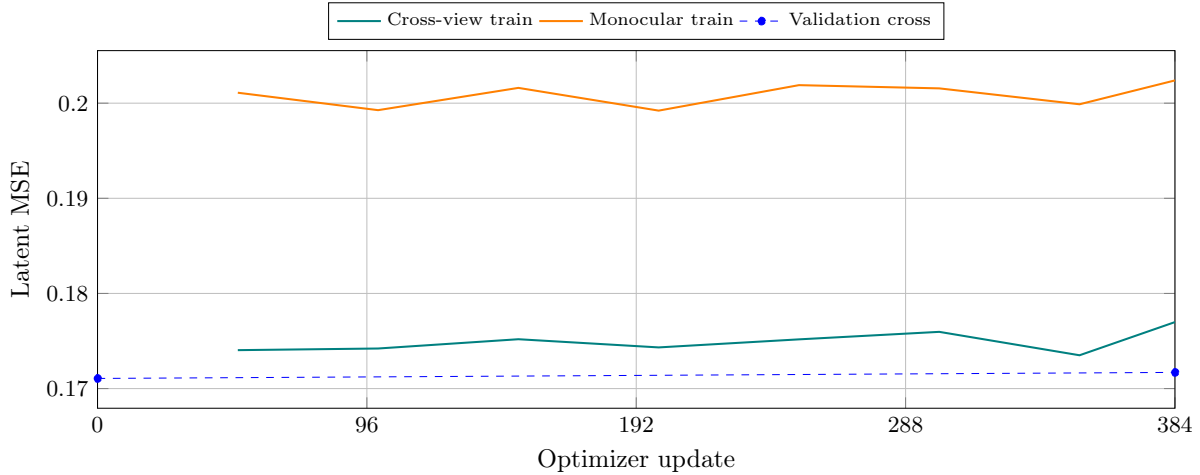
Sparse target tokens retain their original image-grid positions before encoder attention. Reference views are independently encoded. Fusion uses image-grid rotary position encodings and a shared reference role. No camera intrinsics, extrinsics or 3D rays enter this checkpoint. A fixed MIT V-JEPA 2.1 teacher supervises normalized latent targets; dense teacher features are used by the loss only. The full-target relative-improvement branch is evaluated separately from sparse completion.

2 Experiment and data

The selected checkpoint is 0639afa2e19d0e7a. This phase completed 384 updates over 2048 configured cached training rooms, with 6144 logged target exposures covering 2048 unique rooms. Configurations, weight ancestry, dataset identities and the selected checkpoint hash are retained in the bundled machine-readable results. User inputs are TOML; metrics and prediction artifacts are typed native Rust exports. Pretrained noncommercial Gekko weights are excluded from this training lineage.

Completion evaluation role: **development**. Hidden-token MSE averages channels and hidden patches, then target views. Confidence intervals resample whole rooms rather than correlated pixels. The evaluation contains 512 target views. Evaluation geometry is loaded after inference to score co-visibility and correspondence; any training-room geometry supervision is declared separately above. Teacher-fitted display projections are visualization tools and do not enter inference or evaluation metrics.

3 Optimization trajectory



Training curves use nonoverlapping groups of at most 50 updates. Scheduled validation probes are shown when available. Every point belongs to this phase at or before the selected checkpoint. Validation uses development rooms and its declared masks; these curves do not establish independent generalization or training-seed convergence.

4 How to read the metrics

Calibrated camera-motion probe A CPU geometric solver estimates motion from the model’s matches and known intrinsics. Rotation and signed translation-direction errors use degrees; lower is better. Pose error is the larger angle. Pose AUC summarizes recall up to its angular threshold, displayed as a percentage; higher is better. Failed fits count as 180 degrees. Solver success means an estimate was returned, not that it was correct. This is separate from learned camera or focal-length prediction.

Feature signal / error (dB) $10 \log_{10}(\text{mean squared teacher feature} / \text{prediction MSE})$, measured on hidden patches and averaged over target views. Higher is better; +3.01 dB means half the error at the same signal power. It is not RGB PSNR and has no universal image-quality threshold.

RGB PSNR (dB) $10 \log_{10}(\text{RGB data range squared} / \text{RGB MSE})$. Reported only for an evaluated RGB decoder with a declared range, color space and mask. Colored feature plots cannot be scored as reconstructed images.

Feature MSE and cosine MSE is mean squared feature error: zero is exact. Cosine measures feature-direction agreement: one is perfect alignment. A high cosine can coexist with lost spatial detail, so also inspect the variation ratio and shared-color feature maps.

Reference benefit (%) $100 \times (1 - \text{cross-view MSE} / \text{references-disabled MSE})$, using population means from the same checkpoint and targets. Positive means the reference images reduce error. Negative means they hurt.

Spatial variation retained (%) Predicted spatial feature variance divided by teacher variance. 100% matches the teacher’s amount of variation; lower values indicate smoothing. More than 100% is not automatically better, and matched variance does not prove correct structure.

Mean match error (pixels) AEPE: average distance between a predicted correspondence and its true position. Lower is better. HPatches uses the declared 240 by 240 scoring frame; ETH3D uses original image pixels. Neither is measured in the displayed thumbnail’s pixels.

Within 3 pixels (%) PCK3: percentage of labeled matches whose error is at most three pixels in the declared scoring frame. Higher is better; 100% is perfect. A gain of 1 percentage point means one additional match per hundred after the stated macro weighting.

Co-visibility ranking AUROC compares visible and non-visible patches: 0.5 is chance and 1 is perfect. Average precision (AP) also depends on how many patches are visible, so read it alongside the positive-label fraction. These are ranking measures, not calibrated probabilities.

Known-transform loss (nats) Negative log likelihood of the soft correspondence targets, using natural logarithms. Lower is better. Its scale depends on the grid, temperature and label distribution; compare it only within that fixed objective.

Camera and future heads Rotation and translation-direction errors are angles in degrees (lower is better). Focal-length relative error can be shown as a percentage. Pose AUC reports accuracy over an angular threshold range. No camera, depth or RGB capability is claimed until that head is trained and evaluated.

5 Absolute results for the selected checkpoint

5.1 Learned camera calibration head

Head weights trained from scratch on a frozen foundation; Bevy anchor camera: +X right, +Y up, -Z forward; R maps target axes into reference axes; translation is reference-to-target direction in reference axes. Six-dimensional rotation columns, signed unit-baseline regression, log-positive focal lengths. No supplied calibration enters inference.

Metric	Value	Unit / count
Relative rotation error	11.64	degrees / 24
Focal length relative error	44.35	% / 24
Signed translation-direction error	32.68	degrees / 24
Pose AUC at 10 degrees	3.18	% / 24

8 development rooms, 24 target/reference pairs; one training seed. Dense RGB pair input, independent from sparse completion. Principal point fixed at the image center; translation scale is not predicted. 0 invalid rotations; 0 focal clamps.

Training-label constant control: rotation 13.54 degrees, translation 85.29 degrees, focal error 45.64%. Stable regression alone does not prove useful camera estimation.

5.2 Learned RGB reconstruction head

One trained 16x16 patch decoder shared by cross-view and references-disabled latent predictions. sRGB [0,1], data range 1, only originally hidden pixels scored; PSNR computed per view then averaged. Ground-truth pixels are never pasted into scored predictions.

Metric	Value	Unit / count
Hidden-pixel RGB PSNR	21.09	dB / 24
References-disabled RGB PSNR	20.70	dB / 24
Hidden-pixel RGB MSE	0.0089	squared sRGB units / 24

8 development rooms; fixed foundation and one head-training seed. This small head study does not establish sharp completion or real-scene transfer. Colored latent maps elsewhere remain feature visualizations.

5.3 Camera and RGB training stability

PASSED: fixed schedule, independent optimizers, warmup, global gradient clipping at 1 per head, finite-value checks and saved-weight replay. Failed gates: []. Final checkpoint only; no validation selection.

Metric	Value	Unit / count
Maximum camera gradient norm before clipping	4.1301	L2 norm / 800
Maximum RGB gradient norm before clipping	0.0545	L2 norm / 800

Frozen-feature CPU phase: 800 updates in 64.7 seconds. Initial/final validation RGB PSNR 13.46/21.09 dB; camera regression loss 0.6206/0.3874. This verifies bounded optimization, not full encoder/fusion joint-training stability.

5.4 Cross-view latent completion

development; hidden tokens only, fixed V-JEPA 2.1 teacher, equal target-view weighting

Metric	Value	Unit / count
Feature signal / error (higher is better)	7.38	dB / 512
Error reduction from reference views	7.18	% / 512
Hidden-token MSE	0.1843	squared normalized feature units / 512
Hidden-token cosine	0.9026	cosine / 512
Teacher spatial variation retained (100% is reference)	43.56	% / 512

Feature colors visualize latent space; they are not RGB reconstructions.

5.5 Information-use controls

Same checkpoint and masked targets as completion; only available reference information changes. Compare these MSE values with cross-view completion above.

Metric	Value	Unit / count
References disabled	0.1986	MSE / 512
Reference positions shuffled	0.1940	MSE / 512
Unrelated reference room	0.2420	MSE / 512
Training-set mean at each position	0.3084	MSE / 512

Controls measure information use within this experiment; they are not separately trained model versions.

5.6 Completion by reference visibility

Post-hoc descriptive breakdown using geometry loaded after inference. Counts are hidden patches; errors are token-weighted, unlike the primary equal-view aggregate.

Metric	Value	Unit / count
Majority not visible: feature MSE	0.1960	MSE / 26243
Majority visible in references: feature MSE	0.1809	MSE / 91472

A binary majority label does not mean every pixel in the patch is visible or occluded. This is a diagnostic, not a registered acceptance gate or a causal explanation of smoothing.

5.7 Co-visibility

hidden patches only; at least 128 known pixels; majority visible in any reference; ties counted visible; these evaluation co-visibility labels never supervise training. Separate training-room patch-center correspondences supervise the renderer auxiliary; they use a different labeling policy and no evaluation rooms. Geometry is never a model input.

Metric	Value	Unit / count
RI AUROC	0.7382	AUROC / 117715
RI average precision	0.9074	AP / 117715

RI uses a separate full-target branch; it is not sparse completion.

Positive prevalence 77.7%; AP depends on this prevalence.

5.8 HPatches correspondence

development; HP-240; pair then sequence means; primary viewpoint subset; patch-grid displacement

Metric	Value	Unit / count
Pair-conditioned descriptor / Mean match error	15.67	pixels / 295
Pair-conditioned descriptor / Within 3 pixels	15.53	% / 295
Pair-conditioned descriptor + local refinement / Mean match error	14.84	pixels / 295
Pair-conditioned descriptor + local refinement / Within 3 pixels	26.23	% / 295
Same-image conditioned control / Mean match error	19.58	pixels / 295
Same-image conditioned control / Within 3 pixels	13.42	% / 295
Same-image conditioned control + local refinement / Mean match error	18.90	pixels / 295
Same-image conditioned control + local refinement / Within 3 pixels	21.29	% / 295
Encoder block 6 control / Mean match error	19.98	pixels / 295
Encoder block 6 control / Within 3 pixels	13.39	% / 295
Encoder block 6 control + local refinement / Mean match error	19.33	pixels / 295
Encoder block 6 control + local refinement / Within 3 pixels	20.33	% / 295

Local patch-grid readouts; optional 3x3 refinement is declared per method. No parity with published refinement or SOTA claim.

Pair-conditioned descriptor: AEPE 95% cluster interval [13.043, 18.740] pixels over 59 sequence clusters; 10,000 deterministic bootstrap replicates.

Pair-conditioned descriptor + local refinement: AEPE 95% cluster interval [12.190, 17.942] pixels over 59 sequence clusters; 10,000 deterministic bootstrap replicates.

Same-image conditioned control: AEPE 95% cluster interval [16.759, 22.916] pixels over 59 sequence clusters; 10,000 deterministic bootstrap replicates.

Same-image conditioned control + local refinement: AEPE 95% cluster interval [16.042, 22.261] pixels over 59 sequence clusters; 10,000 deterministic bootstrap replicates.

Encoder block 6 control: AEPE 95% cluster interval [17.363, 23.008] pixels over 59 sequence clusters; 10,000 deterministic bootstrap replicates.

Encoder block 6 control + local refinement: AEPE 95% cluster interval [16.682, 22.372] pixels over 59 sequence clusters; 10,000 deterministic bootstrap replicates.

5.9 hpatches: paired readout gate – PASS

Pair-conditioned descriptor relative to Encoder block 6 control; Positive favors candidate: control minus candidate AEPE, candidate minus control PCK3. Pair means, then equal groups within each scene/sequence, then cluster bootstrap. One checkpoint; no training-seed uncertainty.

Metric	Value	Unit / count
Mean pixel error reduction	4.31	pixels / 295
Accuracy gain at 3 pixels	2.1369	percentage points / 295

Paired 95% AEPE-gain interval [3.7844, 4.8579] pixels; PCK3-gain interval [1.831, 2.444] percentage points. 59 clusters, 10,000 deterministic draws.

Fusion transfer gate passes: the pixel-error reduction interval must be entirely positive and mean within-3-pixel accuracy must not decrease. This is not an equivalence test or evidence of general SOTA performance.

5.10 hpatches: paired readout gate – PASS

Pair-conditioned descriptor relative to Same-image conditioned control; Positive favors candidate: control minus candidate AEPE, candidate minus control PCK3. Pair means, then equal groups within each scene/sequence, then cluster bootstrap. One checkpoint; no training-seed uncertainty.

Metric	Value	Unit / count
Mean pixel error reduction	3.91	pixels / 295
Accuracy gain at 3 pixels	2.1120	percentage points / 295

Paired 95% AEPE-gain interval [3.2984, 4.5906] pixels; PCK3-gain interval [1.821, 2.413] percentage points. 59 clusters, 10,000 deterministic draws.

Fusion transfer gate passes: the pixel-error reduction interval must be entirely positive and mean within-3-pixel accuracy must not decrease. This is not an equivalence test or evidence of general SOTA performance.

5.11 hpatches: paired readout gate – PASS

Pair-conditioned descriptor + local refinement relative to Encoder block 6 control + local refinement; Positive favors candidate: control minus candidate AEPE, candidate minus control PCK3. Pair means, then equal groups within each scene/sequence, then cluster bootstrap. One checkpoint; no training-seed uncertainty.

Metric	Value	Unit / count
Mean pixel error reduction	4.49	pixels / 295
Accuracy gain at 3 pixels	5.9061	percentage points / 295

Paired 95% AEPE-gain interval [3.9566, 5.0364] pixels; PCK3-gain interval [5.280, 6.564] percentage points. 59 clusters, 10,000 deterministic draws.

Fusion transfer gate passes: the pixel-error reduction interval must be entirely positive and mean within-3-pixel accuracy must not decrease. This is not an equivalence test or evidence of general SOTA performance.

5.12 hpatches: paired readout gate – PASS

Pair-conditioned descriptor + local refinement relative to Same-image conditioned control + local refinement; Positive favors candidate: control minus candidate AEPE, candidate minus control PCK3. Pair means, then equal groups within each scene/sequence, then cluster bootstrap. One checkpoint; no training-seed uncertainty.

Metric	Value	Unit / count
Mean pixel error reduction	4.06	pixels / 295
Accuracy gain at 3 pixels	4.9392	percentage points / 295

Paired 95% AEPE-gain interval [3.4311, 4.7396] pixels; PCK3-gain interval [4.363, 5.549] percentage points. 59 clusters, 10,000 deterministic draws.

Fusion transfer gate passes: the pixel-error reduction interval must be entirely positive and mean within-3-pixel accuracy must not decrease. This is not an equivalence test or evidence of general SOTA performance.

5.13 hpatches: paired readout gate – PASS

Pair-conditioned descriptor + local refinement relative to Pair-conditioned descriptor; Positive favors candidate: control minus candidate AEPE, candidate minus control PCK3. Pair means, then equal groups within each scene/sequence, then cluster bootstrap. One checkpoint; no training-seed uncertainty.

Metric	Value	Unit / count
Mean pixel error reduction	0.83	pixels / 295
Accuracy gain at 3 pixels	10.7054	percentage points / 295

Paired 95% AEPE-gain interval [0.7636, 0.8993] pixels; PCK3-gain interval [9.489, 12.042] percentage points. 59 clusters, 10,000 deterministic draws.

Local precision gate passes: the within-3-pixel gain interval must be entirely positive and mean pixel error must not increase. This is not an equivalence test or evidence of general SOTA performance.

5.14 ETH3D correspondence

development; original pixels; pair then scene then interval means; patch-grid displacement

Metric	Value	Unit / count
Pair-conditioned descriptor / Mean match error	27.20	pixels / 3365
Pair-conditioned descriptor / Within 3 pixels	2.34	% / 3365
Pair-conditioned descriptor + local refinement / Mean match error	23.81	pixels / 3365
Pair-conditioned descriptor + local refinement / Within 3 pixels	5.68	% / 3365
Same-image conditioned control / Mean match error	34.28	pixels / 3365
Same-image conditioned control / Within 3 pixels	2.19	% / 3365
Same-image conditioned control + local refinement / Mean match error	31.52	pixels / 3365
Same-image conditioned control + local refinement / Within 3 pixels	4.65	% / 3365
Encoder block 6 control / Mean match error	32.96	pixels / 3365
Encoder block 6 control / Within 3 pixels	2.30	% / 3365
Encoder block 6 control + local refinement / Mean match error	30.57	pixels / 3365
Encoder block 6 control + local refinement / Within 3 pixels	4.30	% / 3365

Local patch-grid readouts; optional 3x3 refinement is declared per method. No parity with published refinement or SOTA claim.

Pair-conditioned descriptor: AEPE 95% cluster interval [22.922, 31.869] pixels over 10 scene clusters; 10,000 deterministic bootstrap replicates.

Pair-conditioned descriptor + local refinement: AEPE 95% cluster interval [19.209, 28.804] pixels over 10 scene clusters; 10,000 deterministic bootstrap replicates.

Same-image conditioned control: AEPE 95% cluster interval [29.145, 39.384] pixels over 10 scene clusters; 10,000 deterministic bootstrap replicates.

Same-image conditioned control + local refinement: AEPE 95% cluster interval [26.177, 36.783] pixels over 10 scene clusters; 10,000 deterministic bootstrap replicates.

Encoder block 6 control: AEPE 95% cluster interval [28.351, 37.521] pixels over 10 scene clusters; 10,000 deterministic bootstrap replicates.

Encoder block 6 control + local refinement: AEPE 95% cluster interval [25.862, 35.253] pixels over 10 scene clusters; 10,000 deterministic bootstrap replicates.

5.15 eth3d: paired readout gate – PASS

Pair-conditioned descriptor relative to Encoder block 6 control; Positive favors candidate: control minus candidate AEPE, candidate minus control PCK3. Pair means, then equal groups within each scene/sequence, then cluster bootstrap. One checkpoint; no training-seed uncertainty.

Metric	Value	Unit / count
Mean pixel error reduction	5.76	pixels / 3365
Accuracy gain at 3 pixels	0.0375	percentage points / 3365

Paired 95% AEPE-gain interval [4.3991, 7.3494] pixels; PCK3-gain interval [-0.008, 0.074] percentage points. 10 clusters, 10,000 deterministic draws.

Fusion transfer gate passes: the pixel-error reduction interval must be entirely positive and mean within-3-pixel accuracy must not decrease. This is not an equivalence test or evidence of general SOTA performance.

5.16 eth3d: paired readout gate – PASS

Pair-conditioned descriptor relative to Same-image conditioned control; Positive favors candidate: control minus candidate AEPE, candidate minus control PCK3. Pair means, then equal groups within each scene/sequence, then cluster bootstrap. One checkpoint; no training-seed uncertainty.

Metric	Value	Unit / count
Mean pixel error reduction	7.08	pixels / 3365
Accuracy gain at 3 pixels	0.1497	percentage points / 3365

Paired 95% AEPE-gain interval [4.8046, 9.8022] pixels; PCK3-gain interval [0.106, 0.197] percentage points. 10 clusters, 10,000 deterministic draws.

Fusion transfer gate passes: the pixel-error reduction interval must be entirely positive and mean within-3-pixel accuracy must not decrease. This is not an equivalence test or evidence of general SOTA performance.

5.17 eth3d: paired readout gate – PASS

Pair-conditioned descriptor + local refinement relative to Encoder block 6 control + local refinement; Positive favors candidate: control minus candidate AEPE, candidate minus control PCK3. Pair means, then equal groups within each scene/sequence, then cluster bootstrap. One checkpoint; no training-seed uncertainty.

Metric	Value	Unit / count
Mean pixel error reduction	6.76	pixels / 3365
Accuracy gain at 3 pixels	1.3807	percentage points / 3365

Paired 95% AEPE-gain interval [5.3285, 8.4300] pixels; PCK3-gain interval [1.085, 1.696] percentage points. 10 clusters, 10,000 deterministic draws.

Fusion transfer gate passes: the pixel-error reduction interval must be entirely positive and mean within-3-pixel accuracy must not decrease. This is not an equivalence test or evidence of general SOTA performance.

5.18 eth3d: paired readout gate – PASS

Pair-conditioned descriptor + local refinement relative to Same-image conditioned control + local refinement; Positive favors candidate: control minus candidate AEPE, candidate minus control PCK3. Pair means, then equal groups within each scene/sequence, then cluster bootstrap. One checkpoint; no training-seed uncertainty.

Metric	Value	Unit / count
Mean pixel error reduction	7.71	pixels / 3365
Accuracy gain at 3 pixels	1.0377	percentage points / 3365

Paired 95% AEPE-gain interval [5.3247, 10.5373] pixels; PCK3-gain interval [0.786, 1.293] percentage points. 10 clusters, 10,000 deterministic draws.

Fusion transfer gate passes: the pixel-error reduction interval must be entirely positive and mean within-3-pixel accuracy must not decrease. This is not an equivalence test or evidence of general SOTA performance.

5.19 eth3d: paired readout gate – PASS

Pair-conditioned descriptor + local refinement relative to Pair-conditioned descriptor; Positive favors candidate: control minus candidate AEPE, candidate minus control PCK3. Pair means, then equal groups within each scene/sequence, then cluster bootstrap. One checkpoint; no training-seed uncertainty.

Metric	Value	Unit / count
Mean pixel error reduction	3.39	pixels / 3365
Accuracy gain at 3 pixels	3.3444	percentage points / 3365

Paired 95% AEPE-gain interval [2.9974, 3.7744] pixels; PCK3-gain interval [2.806, 3.998] percentage points. 10 clusters, 10,000 deterministic draws.

Local precision gate passes: the within-3-pixel gain interval must be entirely positive and mean pixel error must not increase. This is not an equivalence test or evidence of general SOTA performance.

5.20 Synthetic camera-view matching

Fixed validation rooms, view 0 versus view 1, both directions. RGB-only predictions; renderer depth and camera labels enter scoring only. Pair, same-image and encoder controls share the query population.

Metric	Value	Unit / count
Centered encoder control / AEPE	20.30	pixels / 32
Centered encoder control / PCK8	34.26	% / 32
Centered encoder control / PCK16	63.32	% / 32
Centered encoder control + local refinement / AEPE	18.81	pixels / 32
Centered encoder control + local refinement / PCK8	41.98	% / 32
Centered encoder control + local refinement / PCK16	67.20	% / 32
Pair-conditioned descriptor / AEPE	14.24	pixels / 32
Pair-conditioned descriptor / PCK8	43.97	% / 32
Pair-conditioned descriptor / PCK16	76.75	% / 32
Pair-conditioned descriptor + local refinement / AEPE	12.18	pixels / 32
Pair-conditioned descriptor + local refinement / PCK8	58.72	% / 32
Pair-conditioned descriptor + local refinement / PCK16	80.21	% / 32

Metrics continued.

Metric	Value	Unit / count
Same-image conditioned control / AEPE	19.56	pixels / 32
Same-image conditioned control / PCK8	36.49	% / 32
Same-image conditioned control / PCK16	65.61	% / 32
Same-image conditioned control + local refinement / AEPE	17.83	pixels / 32
Same-image conditioned control + local refinement / PCK8	46.01	% / 32
Same-image conditioned control + local refinement / PCK16	69.84	% / 32

This is synthetic development evaluation. Training may use renderer-supervised correspondence labels; external real-image transfer is reported separately. Unknown, occluded, out-of-frame and grid-border queries are excluded under the pinned policy.

5.21 Calibrated motion / Pair-conditioned descriptor + local refinement

Evaluation role: development. TUM Freiburg 3 calibrated relative-motion diagnostic. RGB-only 256px model, fixed local centroid and mutual matches; known intrinsics supplied only to deterministic CPU eight-point essential RANSAC. Equal pair weight within sequence, then equal sequences. Failed fits contribute 180 degrees; ground-truth baselines below 1 cm excluded from translation/pose only. Not a learned camera head or official SLAM/relative-pose protocol.

Metric	Value	Unit / count
Solver success	98.92	% / 186
Mean rotation error	7.78	degrees / 186
Mean signed translation-direction error	52.80	degrees / 185
Pose AUC at 5 degrees	1.54	% / 185
Pose AUC at 10 degrees	7.65	% / 185
Pose AUC at 20 degrees	18.86	% / 185
Poses within 10 degrees	21.10	% / 185

Known RGB intrinsics enter a CPU eight-point RANSAC solver after RGB-only model inference. This is not a trained camera head or predicted calibration.

Three sequences in one TUM environment; no official relative-pose benchmark parity or broad generalization claim. Failure counts and low-baseline exclusions remain explicit.

5.22 Calibrated motion / Same-image conditioned control + local refinement

Evaluation role: development. TUM Freiburg 3 calibrated relative-motion diagnostic. RGB-only 256px model, fixed local centroid and mutual matches; known intrinsics supplied only to deterministic CPU eight-point

essential RANSAC. Equal pair weight within sequence, then equal sequences. Failed fits contribute 180 degrees; ground-truth baselines below 1 cm excluded from translation/pose only. Not a learned camera head or official SLAM/relative-pose protocol.

Metric	Value	Unit / count
Solver success	98.39	% / 186
Mean rotation error	9.50	degrees / 186
Mean signed translation-direction error	55.15	degrees / 185
Pose AUC at 5 degrees	1.79	% / 185
Pose AUC at 10 degrees	6.54	% / 185
Pose AUC at 20 degrees	18.16	% / 185
Poses within 10 degrees	17.35	% / 185

Known RGB intrinsics enter a CPU eight-point RANSAC solver after RGB-only model inference. This is not a trained camera head or predicted calibration.

Three sequences in one TUM environment; no official relative-pose benchmark parity or broad generalization claim. Failure counts and low-baseline exclusions remain explicit.

5.23 Calibrated motion / Encoder block 6 control + local refinement

Evaluation role: development. TUM Freiburg 3 calibrated relative-motion diagnostic. RGB-only 256px model, fixed local centroid and mutual matches; known intrinsics supplied only to deterministic CPU eight-point essential RANSAC. Equal pair weight within sequence, then equal sequences. Failed fits contribute 180 degrees; ground-truth baselines below 1 cm excluded from translation/pose only. Not a learned camera head or official SLAM/relative-pose protocol.

Metric	Value	Unit / count
Solver success	98.92	% / 186
Mean rotation error	8.78	degrees / 186
Mean signed translation-direction error	56.31	degrees / 185
Pose AUC at 5 degrees	1.78	% / 185
Pose AUC at 10 degrees	5.85	% / 185
Pose AUC at 20 degrees	16.92	% / 185
Poses within 10 degrees	16.79	% / 185

Known RGB intrinsics enter a CPU eight-point RANSAC solver after RGB-only model inference. This is not a trained camera head or predicted calibration.

Three sequences in one TUM environment; no official relative-pose benchmark parity or broad generalization claim. Failure counts and low-baseline exclusions remain explicit.

5.24 Camera-motion transfer gate

FAILED: AUC@10 must improve over both controls in every sequence, without lower overall solver success.

Metric	Value	Unit / count
Pose AUC@10 gain over Same-image conditioned control + local refinement	1.1072	percentage points / 3
Pose AUC@10 gain over Encoder block 6 control + local refinement	1.8020	percentage points / 3

Only three related sequences; the gate is a bounded diagnostic, not a SOTA qualification.

5.25 Depth / future decoders

Register a checkpoint-bound head evaluation to populate this capability. **No evaluated capability is claimed for this head.** Completion room-bootstrap 95% interval: [0.181129, 0.187492], 128 rooms, 10,000 deterministic replicates. This measures scene sampling uncertainty, not variation across training seeds. Paired gain from using references (monocular MSE minus cross-view MSE): 0.014259, room-bootstrap 95% interval [0.012480, 0.016027]. Positive values favor reference use.

5.26 Camera motion by sequence

TUM RGB-D data [9] is CC BY 4.0. These three sequences share one acquisition environment; prior exposure in encoder pretraining is not established. Known intrinsics enter the CPU solver only. Solver success counts estimates returned, including inaccurate ones. Failed fits contribute 180 degrees. Exclusions are baselines below 1 cm and affect translation/pose only.

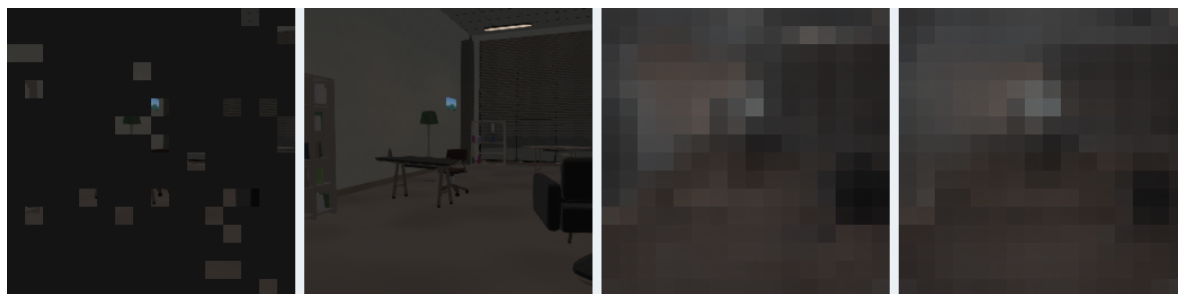
Sequence	Readout	Solved/all	Rot. °	Trans. °	AUC@10 %	Excl.
Office	Fusion	60/62	13.13	50.19	6.82	0
Structure far	Fusion	62/62	4.73	53.58	7.90	0
Structure near	Fusion	62/62	5.47	54.62	8.23	1
Office	Same-image	60/62	14.97	63.47	3.77	0
Structure far	Same-image	62/62	4.57	56.32	5.92	0
Structure near	Same-image	61/62	8.96	45.67	9.93	1
Office	Encoder	60/62	15.20	59.83	4.62	0
Structure far	Encoder	62/62	4.76	62.20	6.22	0
Structure near	Encoder	62/62	6.37	46.91	6.71	1

6 Visual evaluation protocol

Examples are evenly spaced over the exported sample identities, retaining the first and last. They are not selected by prediction quality. All displayed teacher features share one three-component PCA projection and teacher-derived 2nd–98th percentile color bounds; predictions use the identical projection and bounds. Color structure describes representation variation, not RGB texture reconstruction. Error maps use a fixed 0–1 scale (dark blue to teal to yellow to red), saturating above 1. Co-visibility truth uses teal for shared geometry, coral for non-visible patches and gray for observed/unknown patches. Camera-motion panels identify predicted matches and RANSAC inliers; these are not ground-truth correspondences.

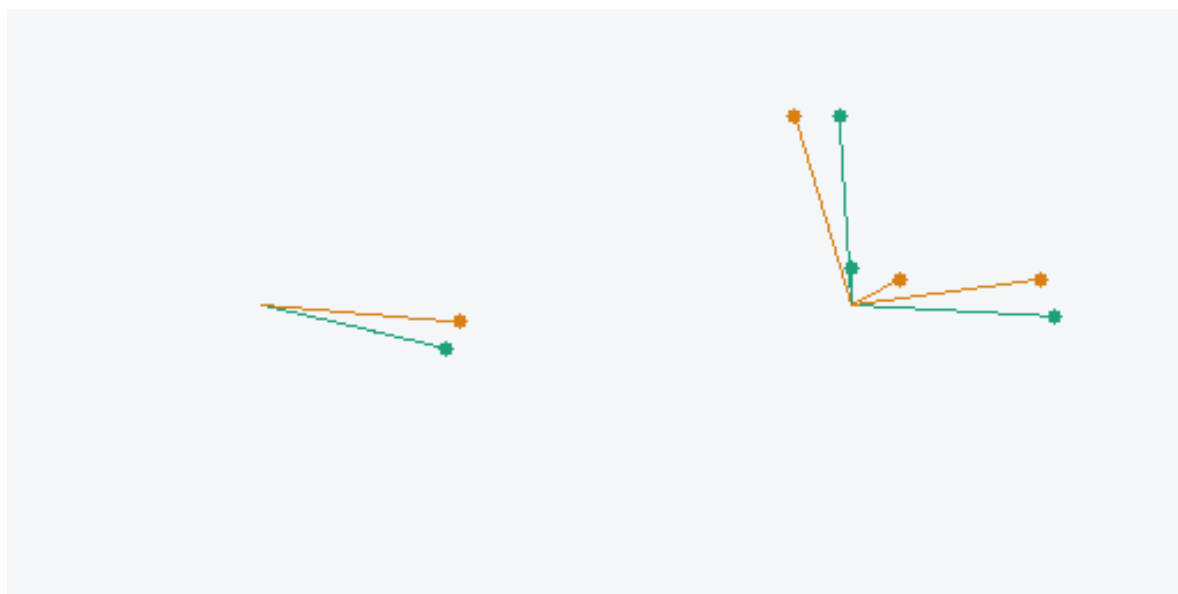
Learned RGB and camera examples

RGB head / room 2609308192, view 0



Left to right: sparse target input, target RGB (evaluation only), predicted RGB, references-disabled RGB. Hidden-pixel PSNR 26.35 / 25.75 dB; the full images shown are generated by the same learned head. These fixed first/middle/last samples are not selected by quality.

Camera head / room 2609308192, view 0



Green: renderer truth; amber: prediction. Left: signed unit translation; right: target orientation axes in the reference frame, under one fixed oblique projection. This is a dense RGB-pair route. Rotation error 19.15 degrees; translation-direction error 13.67 degrees; focal relative error 46.88%. Diagram lengths are not metric translation scale.

Learned RGB and camera examples

RGB head / room 2609308196, view 0



Left to right: sparse target input, target RGB (evaluation only), predicted RGB, references-disabled RGB. Hidden-pixel PSNR 18.61 / 18.38 dB; the full images shown are generated by the same learned head. These fixed first/middle/last samples are not selected by quality.

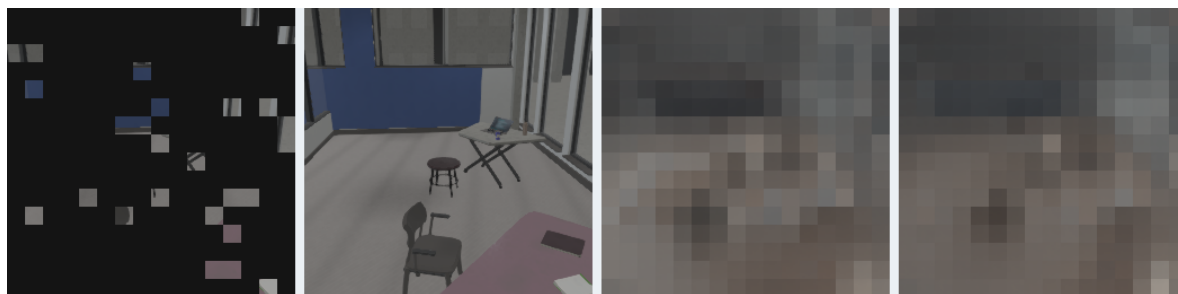
Camera head / room 2609308196, view 0



Green: renderer truth; amber: prediction. Left: signed unit translation; right: target orientation axes in the reference frame, under one fixed oblique projection. This is a dense RGB-pair route. Rotation error 4.75 degrees; translation-direction error 4.33 degrees; focal relative error 35.10%. Diagram lengths are not metric translation scale.

Learned RGB and camera examples

RGB head / room 2609308199, view 2



Left to right: sparse target input, target RGB (evaluation only), predicted RGB, references-disabled RGB. Hidden-pixel PSNR 21.39 / 20.59 dB; the full images shown are generated by the same learned head. These fixed first/middle/last samples are not selected by quality.

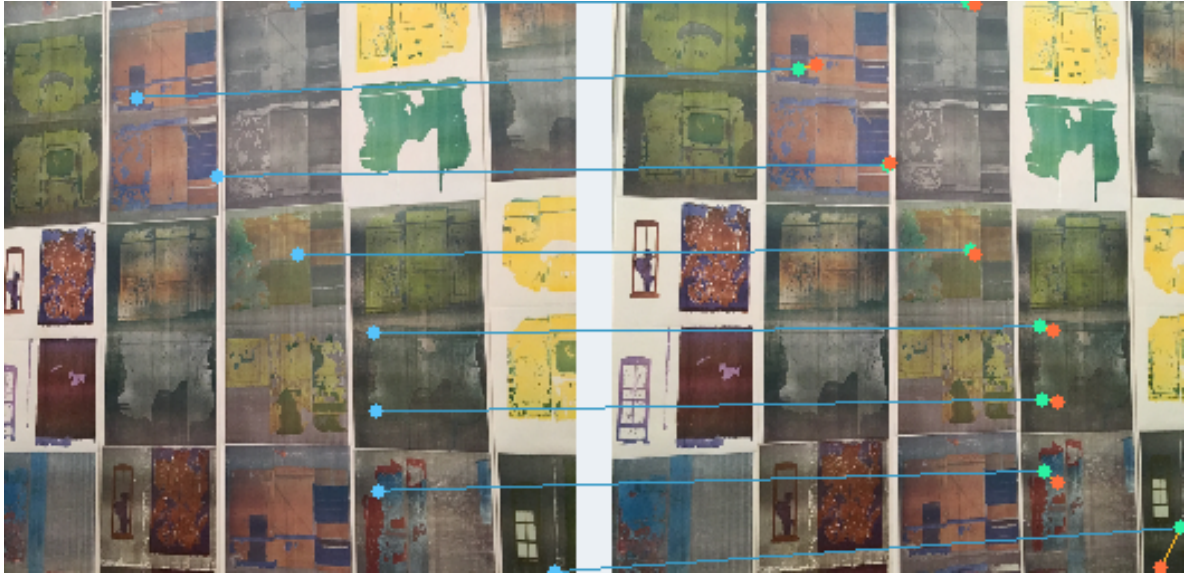
Camera head / room 2609308199, view 2



Green: renderer truth; amber: prediction. Left: signed unit translation; right: target orientation axes in the reference frame, under one fixed oblique projection. This is a dense RGB-pair route. Rotation error 19.98 degrees; translation-direction error 36.09 degrees; focal relative error 22.93%. Diagram lengths are not metric translation scale.

Geometric correspondence examples

hpatches / v_abstract_2



Target left; reference right. Blue: target/true-match connection. Green: ground-truth reference location. Orange: model match. Yellow: matching error. Pair-conditioned descriptor + local refinement readout; mean match error 4.14 pixels in the 240 by 240 scoring frame. Eight labels selected uniformly, not by error. first/middle/last lexicographic pair in the primary subset; eight evenly spaced valid labels, no quality filtering.

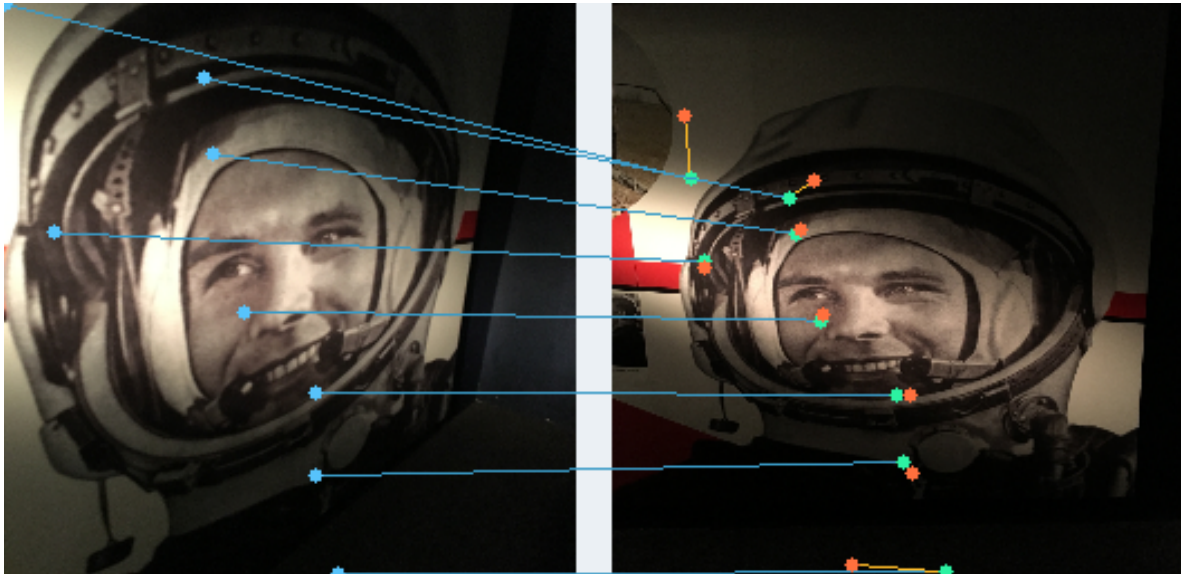
hpatches / v_gardens_4



Target left; reference right. Blue: target/true-match connection. Green: ground-truth reference location. Orange: model match. Yellow: matching error. Pair-conditioned descriptor + local refinement readout; mean match error 4.70 pixels in the 240 by 240 scoring frame. Eight labels selected uniformly, not by error. first/middle/last lexicographic pair in the primary subset; eight evenly spaced valid labels, no quality filtering.

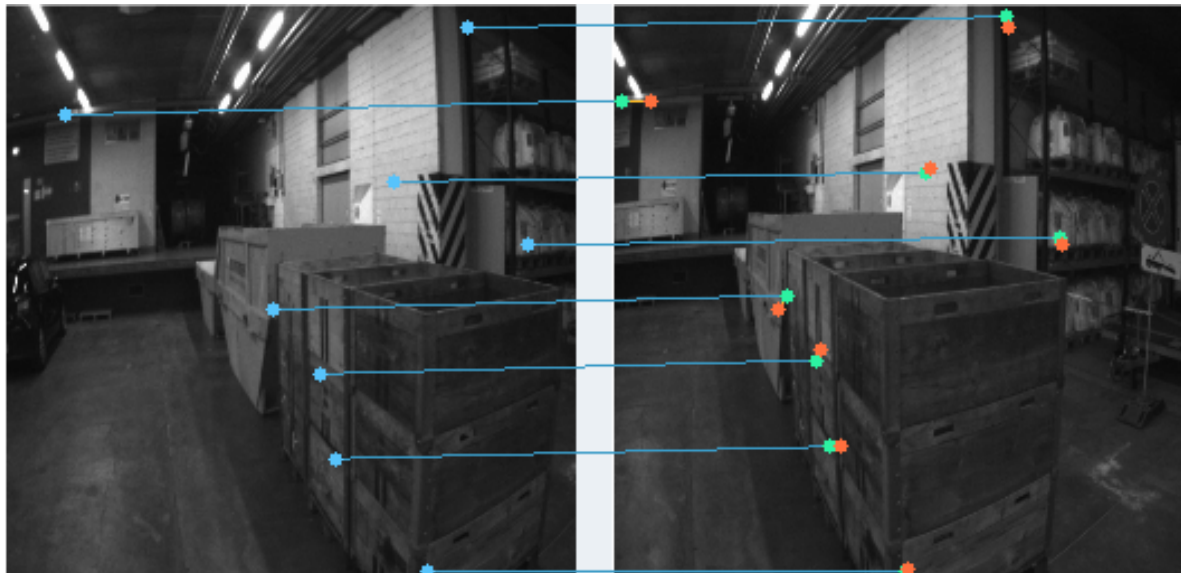
Geometric correspondence examples

hpatches / v_yuri_6



Target left; reference right. Blue: target/true-match connection. Green: ground-truth reference location. Orange: model match. Yellow: matching error. Pair-conditioned descriptor + local refinement readout; mean match error 8.62 pixels in the 240 by 240 scoring frame. Eight labels selected uniformly, not by error. first/middle/last lexicographic pair in the primary subset; eight evenly spaced valid labels, no quality filtering.

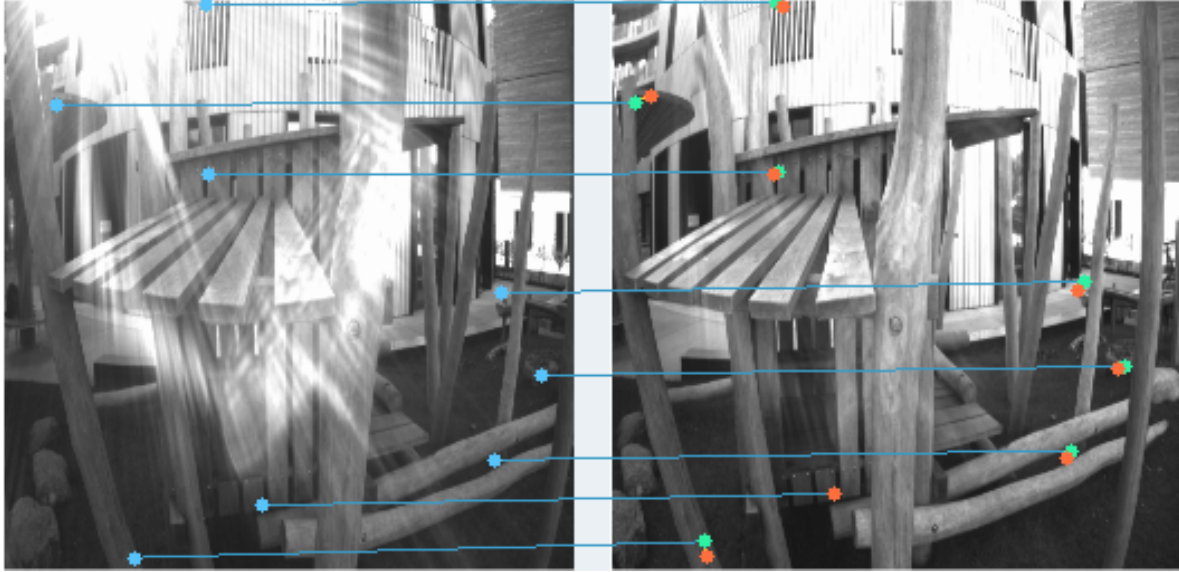
eth3d / delivery_area-11-0000



Target left; reference right. Blue: target/true-match connection. Green: ground-truth reference location. Orange: model match. Yellow: matching error. Pair-conditioned descriptor + local refinement readout; mean match error 11.44 pixels in the 752 by 480 scoring frame. Eight labels selected uniformly, not by error. first/middle/last lexicographic pair in the primary subset; eight evenly spaced valid labels, no quality filtering.

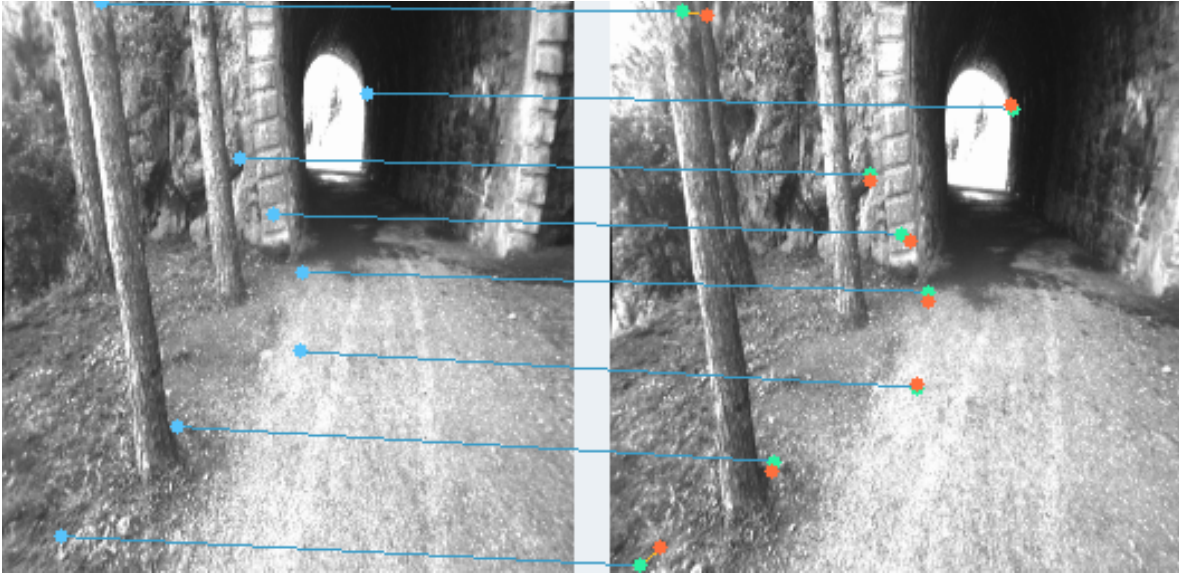
Geometric correspondence examples

eth3d / playground-7-0015



Target left; reference right. Blue: target/true-match connection. Green: ground-truth reference location. Orange: model match. Yellow: matching error. Pair-conditioned descriptor + local refinement readout; mean match error 12.48 pixels in the 752 by 480 scoring frame. Eight labels selected uniformly, not by error. first/middle/last lexicographic pair in the primary subset; eight evenly spaced valid labels, no quality filtering.

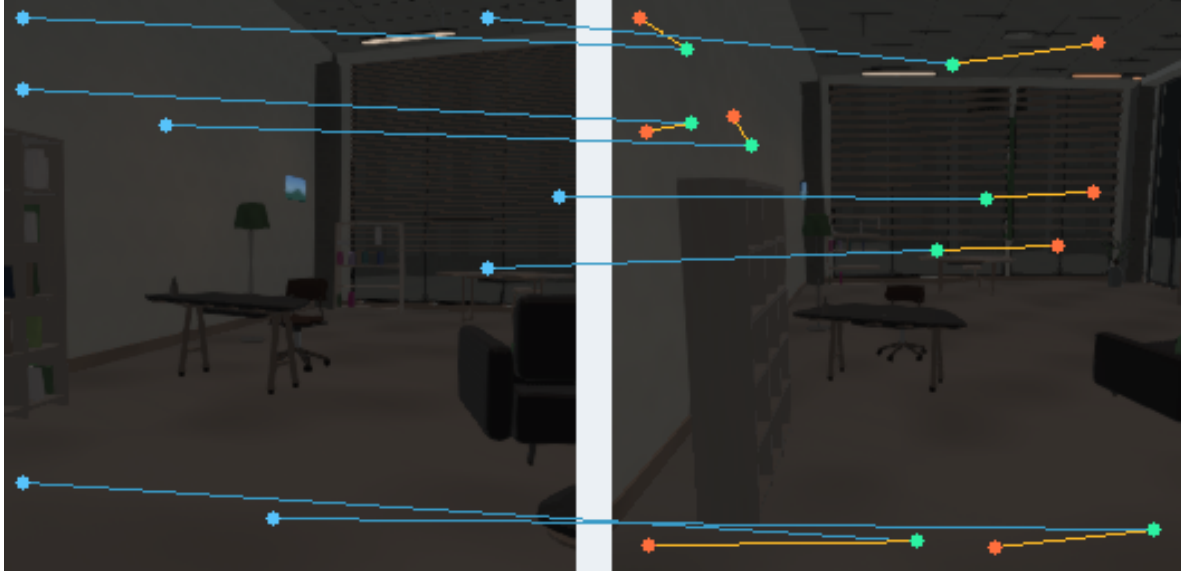
eth3d / tunnel-9-0068



Target left; reference right. Blue: target/true-match connection. Green: ground-truth reference location. Orange: model match. Yellow: matching error. Pair-conditioned descriptor + local refinement readout; mean match error 17.15 pixels in the 957 by 517 scoring frame. Eight labels selected uniformly, not by error. first/middle/last lexicographic pair in the primary subset; eight evenly spaced valid labels, no quality filtering.

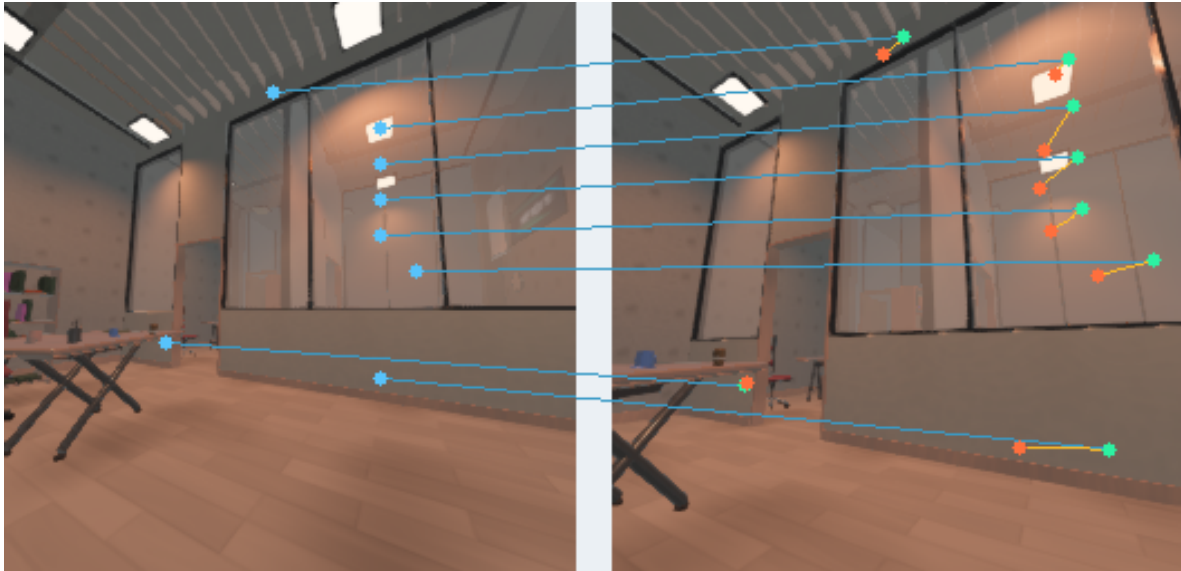
Geometric correspondence examples

Actual camera views / room 2609308192



View 0 left, view 1 right. Blue lines join source queries to renderer-projected green targets; orange marks RGB-model predictions and yellow shows error. Mean endpoint error 25.47 input pixels over all known visible queries. First four validation rooms; eight evenly spaced valid queries. Synthetic development data, not external transfer evidence.

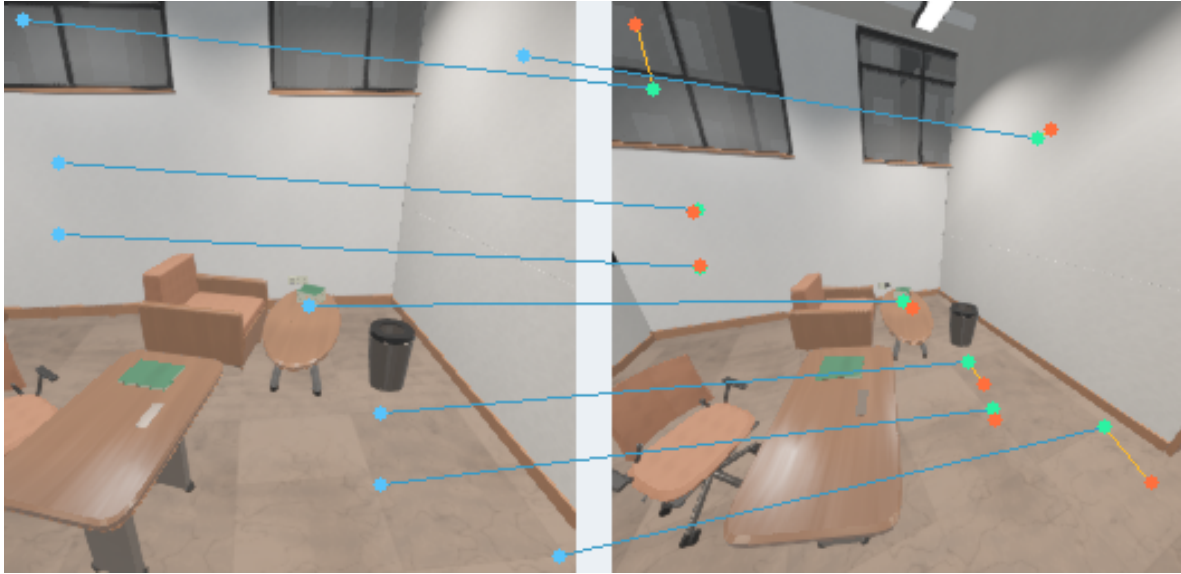
Actual camera views / room 2609308193



View 0 left, view 1 right. Blue lines join source queries to renderer-projected green targets; orange marks RGB-model predictions and yellow shows error. Mean endpoint error 11.08 input pixels over all known visible queries. First four validation rooms; eight evenly spaced valid queries. Synthetic development data, not external transfer evidence.

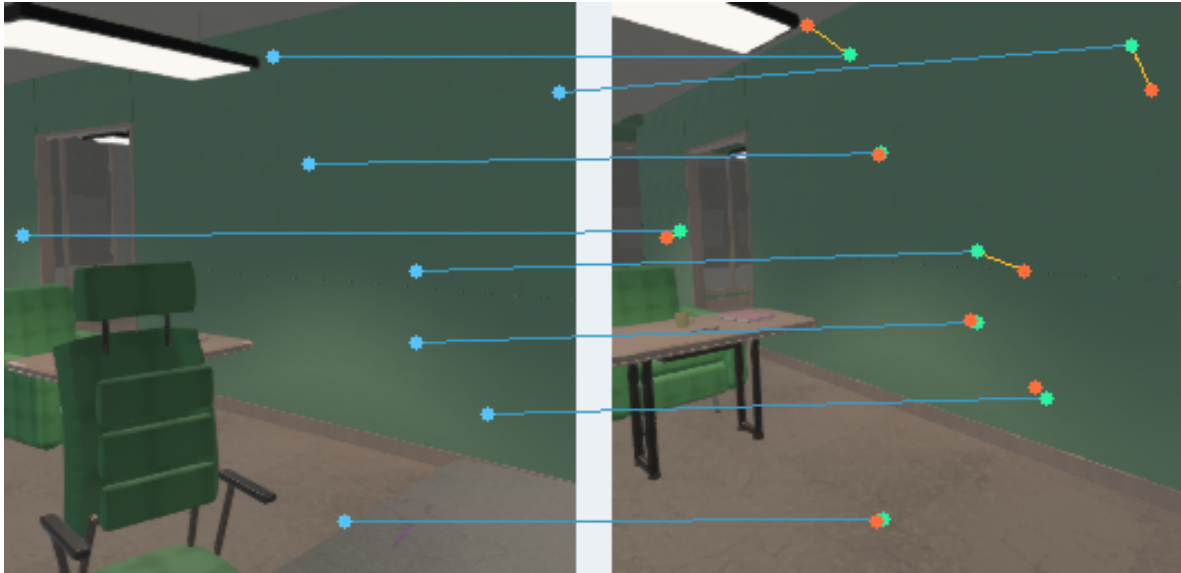
Geometric correspondence examples

Actual camera views / room 2609308194



View 0 left, view 1 right. Blue lines join source queries to renderer-projected green targets; orange marks RGB-model predictions and yellow shows error. Mean endpoint error 11.00 input pixels over all known visible queries. First four validation rooms; eight evenly spaced valid queries. Synthetic development data, not external transfer evidence.

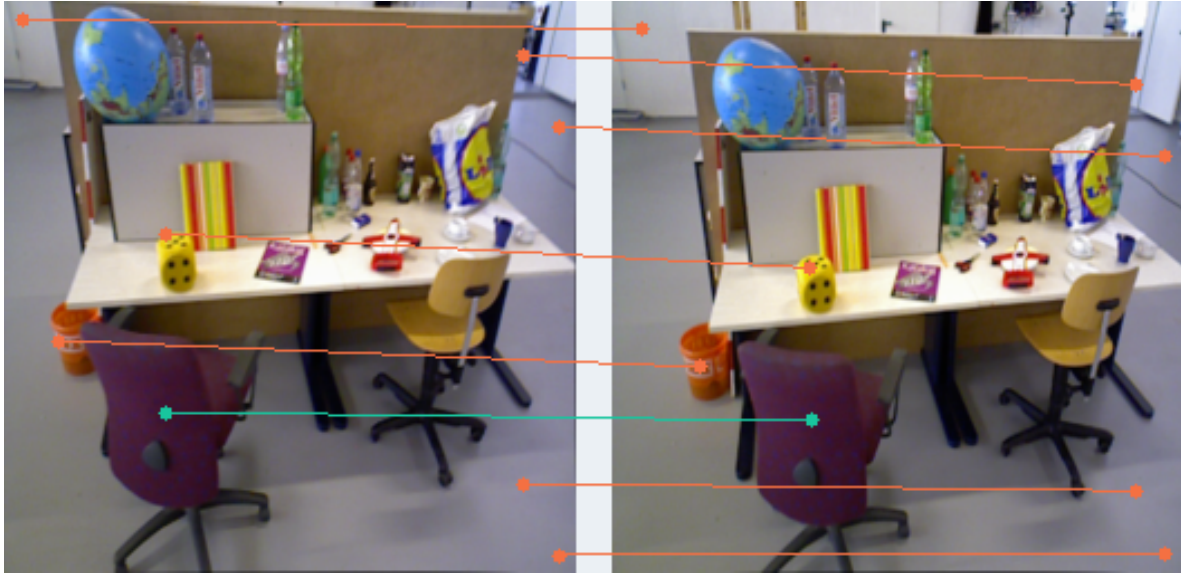
Actual camera views / room 2609308195



View 0 left, view 1 right. Blue lines join source queries to renderer-projected green targets; orange marks RGB-model predictions and yellow shows error. Mean endpoint error 9.08 input pixels over all known visible queries. First four validation rooms; eight evenly spaced valid queries. Synthetic development data, not external transfer evidence.

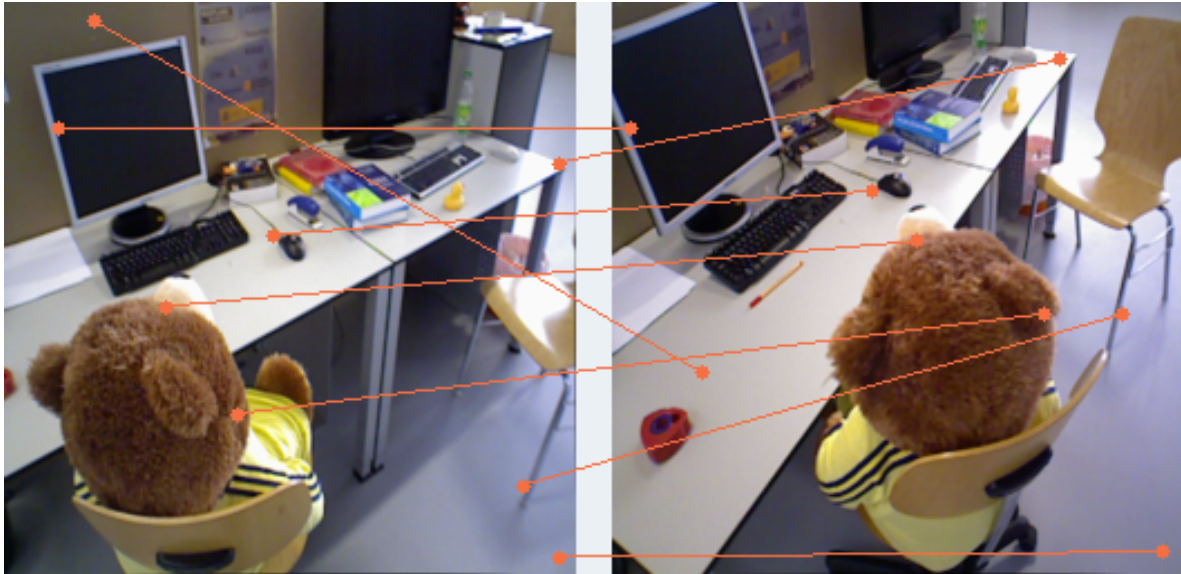
Geometric correspondence examples

TUM camera motion / long_office_household-000081-000096



Target left; reference right. Model RGB input 256px; display thumbnails 256px. Predicted mutual correspondences: green marks RANSAC inliers, orange rejected matches. Inlier agreement is not ground-truth correctness. solver returned a pose; rotation error 3.20 degrees, signed translation-direction error 41.60 degrees. Known intrinsics enter the CPU solver only. first/middle/last lexicographic pair per sequence; up to eight uniformly spaced mutual matches, no quality filtering.

TUM camera motion / long_office_household-001302-001362



Target left; reference right. Model RGB input 256px; display thumbnails 256px. Predicted mutual correspondences: green marks RANSAC inliers, orange rejected matches. Inlier agreement is not ground-truth correctness. solver returned a pose; rotation error 11.44 degrees, signed translation-direction error 21.82 degrees. Known intrinsics enter the CPU solver only. first/middle/last lexicographic pair per sequence; up to eight uniformly spaced mutual matches, no quality filtering.

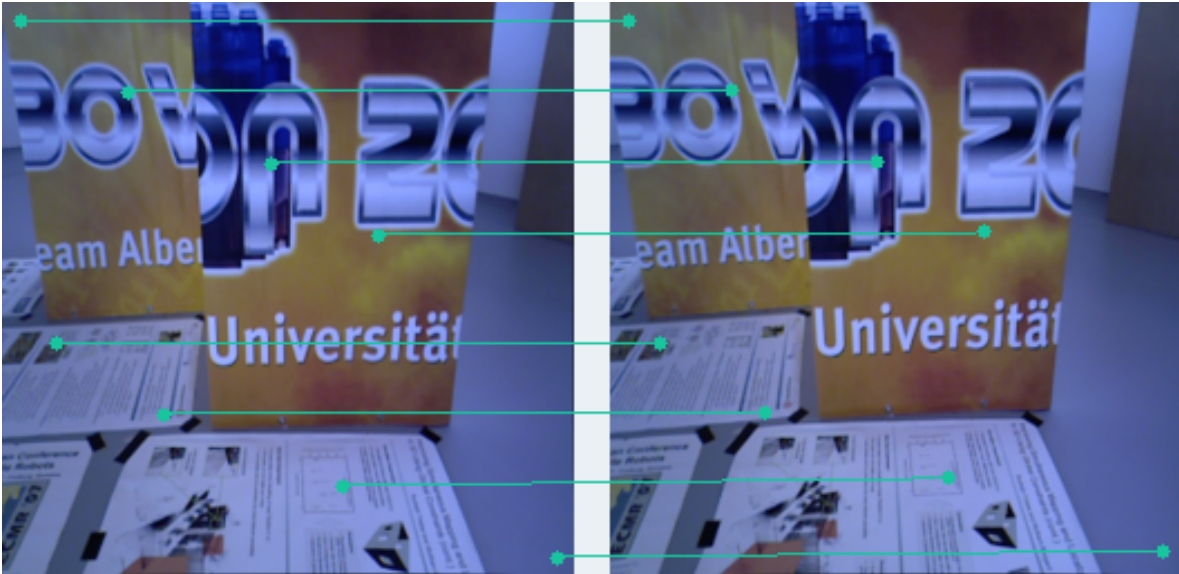
Geometric correspondence examples

TUM camera motion / long_office_household-002524-002584



Target left; reference right. Model RGB input 256px; display thumbnails 256px. Predicted mutual correspondences: green marks RANSAC inliers, orange rejected matches. Inlier agreement is not ground-truth correctness. solver returned a pose; rotation error 2.16 degrees, signed translation-direction error 12.78 degrees. Known intrinsics enter the CPU solver only. first/middle/last lexicographic pair per sequence; up to eight uniformly spaced mutual matches, no quality filtering.

TUM camera motion / structure_texture_far-000028-000043



Target left; reference right. Model RGB input 256px; display thumbnails 256px. Predicted mutual correspondences: green marks RANSAC inliers, orange rejected matches. Inlier agreement is not ground-truth correctness. solver returned a pose; rotation error 1.70 degrees, signed translation-direction error 35.23 degrees. Known intrinsics enter the CPU solver only. first/middle/last lexicographic pair per sequence; up to eight uniformly spaced mutual matches, no quality filtering.

Geometric correspondence examples

TUM camera motion / structure_texture_far-000452-000512



Target left; reference right. Model RGB input 256px; display thumbnails 256px. Predicted mutual correspondences: green marks RANSAC inliers, orange rejected matches. Inlier agreement is not ground-truth correctness. solver returned a pose; rotation error 10.61 degrees, signed translation-direction error 60.88 degrees. Known intrinsics enter the CPU solver only. first/middle/last lexicographic pair per sequence; up to eight uniformly spaced mutual matches, no quality filtering.

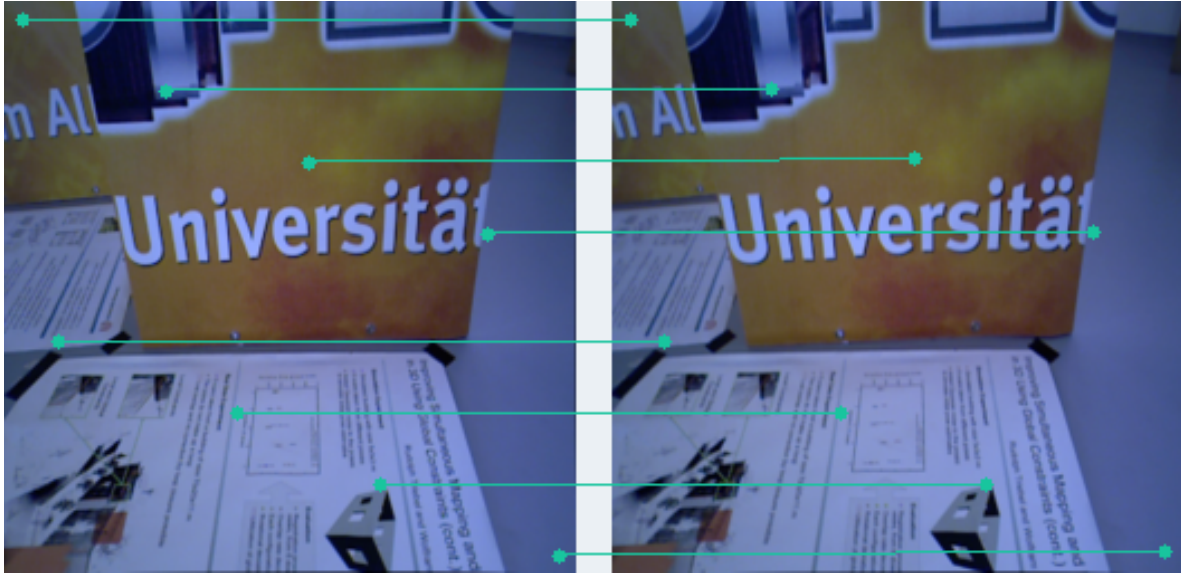
TUM camera motion / structure_texture_far-000877-000937



Target left; reference right. Model RGB input 256px; display thumbnails 256px. Predicted mutual correspondences: green marks RANSAC inliers, orange rejected matches. Inlier agreement is not ground-truth correctness. solver returned a pose; rotation error 3.50 degrees, signed translation-direction error 5.31 degrees. Known intrinsics enter the CPU solver only. first/middle/last lexicographic pair per sequence; up to eight uniformly spaced mutual matches, no quality filtering.

Geometric correspondence examples

TUM camera motion / structure_texture_near-000033-000048



Target left; reference right. Model RGB input 256px; display thumbnails 256px. Predicted mutual correspondences: green marks RANSAC inliers, orange rejected matches. Inlier agreement is not ground-truth correctness. solver returned a pose; rotation error 0.50 degrees, signed translation-direction error excluded: baseline below 1 cm. Known intrinsics enter the CPU solver only. first/middle/last lexicographic pair per sequence; up to eight uniformly spaced mutual matches, no quality filtering.

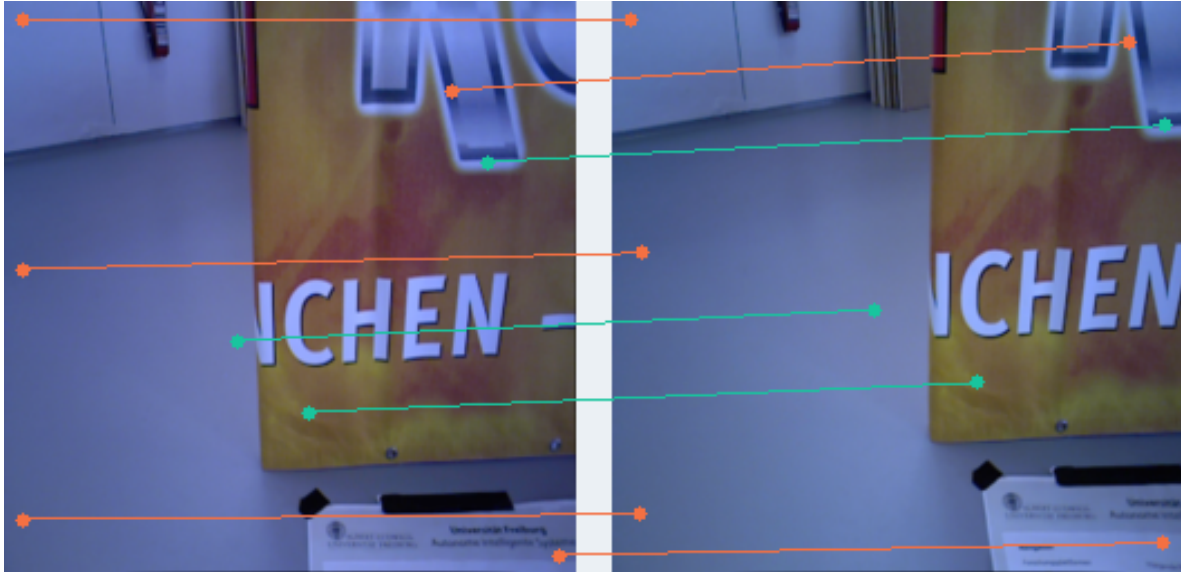
TUM camera motion / structure_texture_near-000535-000595



Target left; reference right. Model RGB input 256px; display thumbnails 256px. Predicted mutual correspondences: green marks RANSAC inliers, orange rejected matches. Inlier agreement is not ground-truth correctness. solver returned a pose; rotation error 22.02 degrees, signed translation-direction error 21.03 degrees. Known intrinsics enter the CPU solver only. first/middle/last lexicographic pair per sequence; up to eight uniformly spaced mutual matches, no quality filtering.

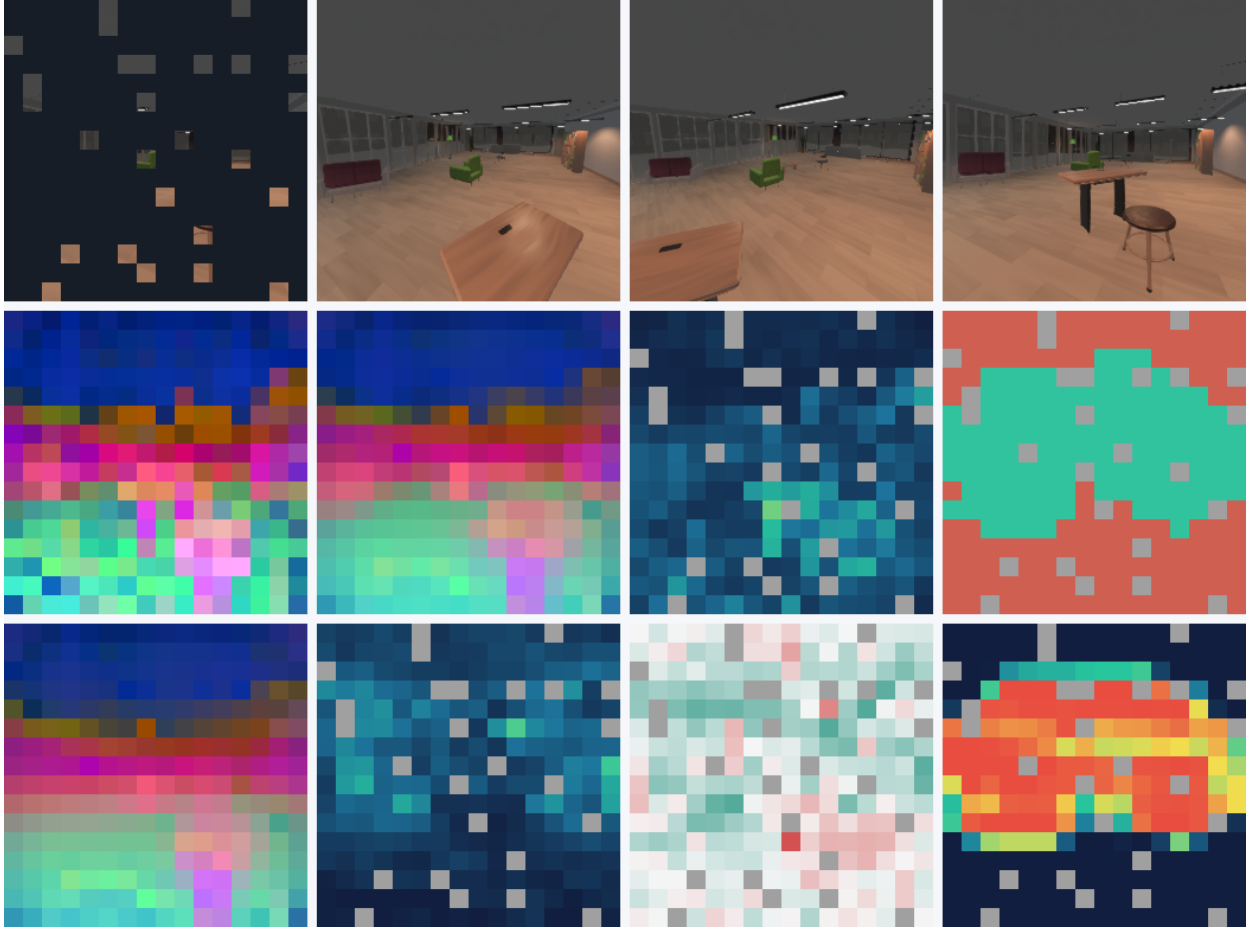
Geometric correspondence examples

TUM camera motion / structure_texture_near-001038-001098



Target left; reference right. Model RGB input 256px; display thumbnails 256px. Predicted mutual correspondences: green marks RANSAC inliers, orange rejected matches. Inlier agreement is not ground-truth correctness. solver returned a pose; rotation error 4.66 degrees, signed translation-direction error 39.50 degrees. Known intrinsics enter the CPU solver only. first/middle/last lexicographic pair per sequence; up to eight uniformly spaced mutual matches, no quality filtering.

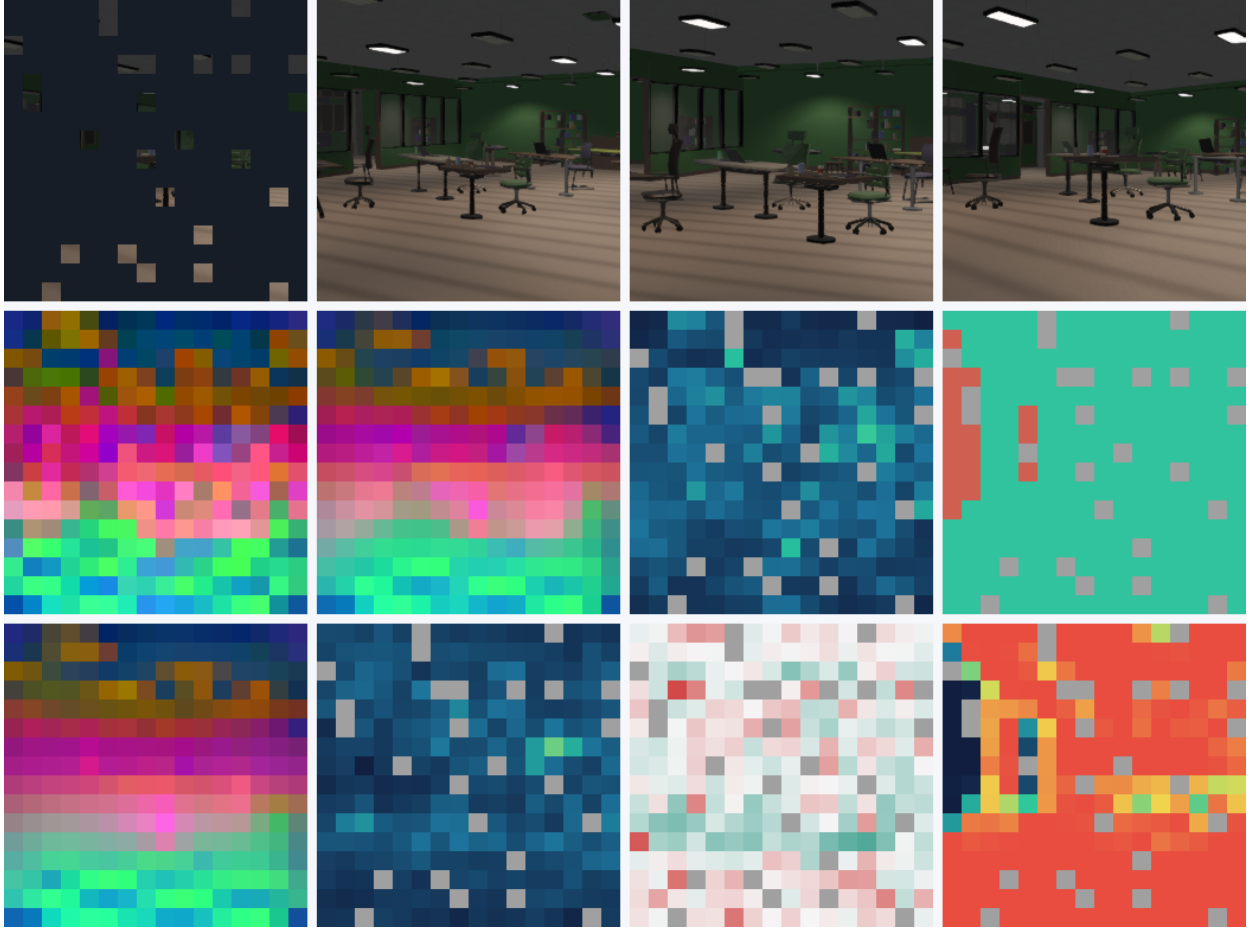
6.1 Room 2610091002, target view 0



Top row, left to right: sparse target input; reference 1; reference 2 (or an explicitly blank panel); full target RGB for evaluation only. Second row: teacher latent; predicted latent; hidden-token error; geometric co-visibility truth. If present, the third row shows the same model's monocular latent, learned RI score, signed reference benefit and geometric visibility fraction. RI/fraction maps use a 0–1 scale. Reference benefit is (monocular MSE minus cross-view MSE) divided by monocular MSE: -1 red (worse), 0 white, +1 teal (better), saturating outside that range. Unlike the clipped training RI target, this display retains negative effects. RI is an unconstrained regression score, not a calibrated visibility probability. The teacher and full target do not enter sparse inference.

Feature signal/error: 8.66 dB (not RGB PSNR). Hidden-token MSE: 0.136011. Cosine: 0.928703. Values are recomputed in Rust from the exported arrays and checked against the population evaluation. All samples use the same error and projection conventions.

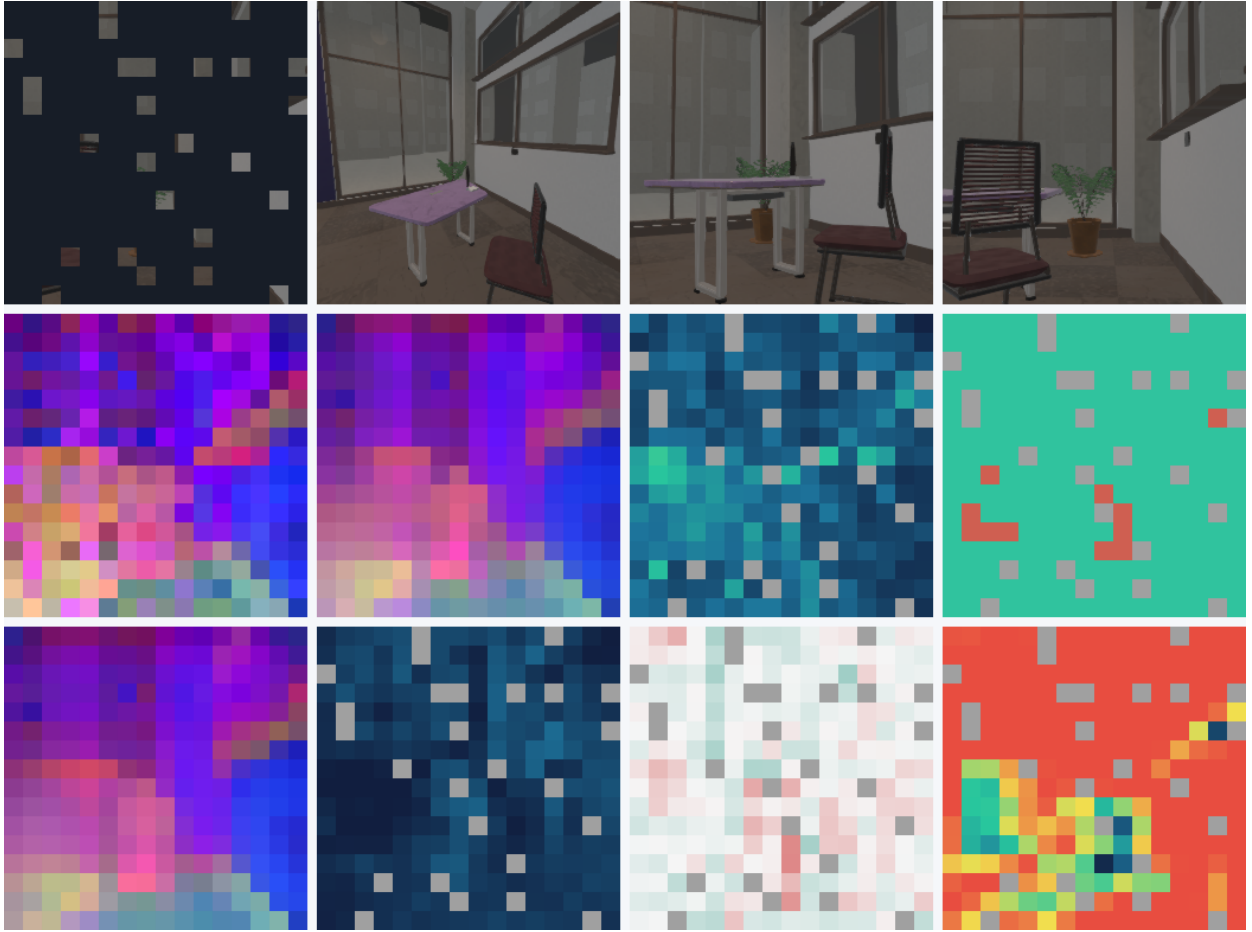
6.2 Room 2610091027, target view 2



Top row, left to right: sparse target input; reference 1; reference 2 (or an explicitly blank panel); full target RGB for evaluation only. Second row: teacher latent; predicted latent; hidden-token error; geometric co-visibility truth. If present, the third row shows the same model's monocular latent, learned RI score, signed reference benefit and geometric visibility fraction. RI/fraction maps use a 0–1 scale. Reference benefit is (monocular MSE minus cross-view MSE) divided by monocular MSE: -1 red (worse), 0 white, +1 teal (better), saturating outside that range. Unlike the clipped training RI target, this display retains negative effects. RI is an unconstrained regression score, not a calibrated visibility probability. The teacher and full target do not enter sparse inference.

Feature signal/error: 7.69 dB (not RGB PSNR). Hidden-token MSE: 0.170213. Cosine: 0.910693. Values are recomputed in Rust from the exported arrays and checked against the population evaluation. All samples use the same error and projection conventions.

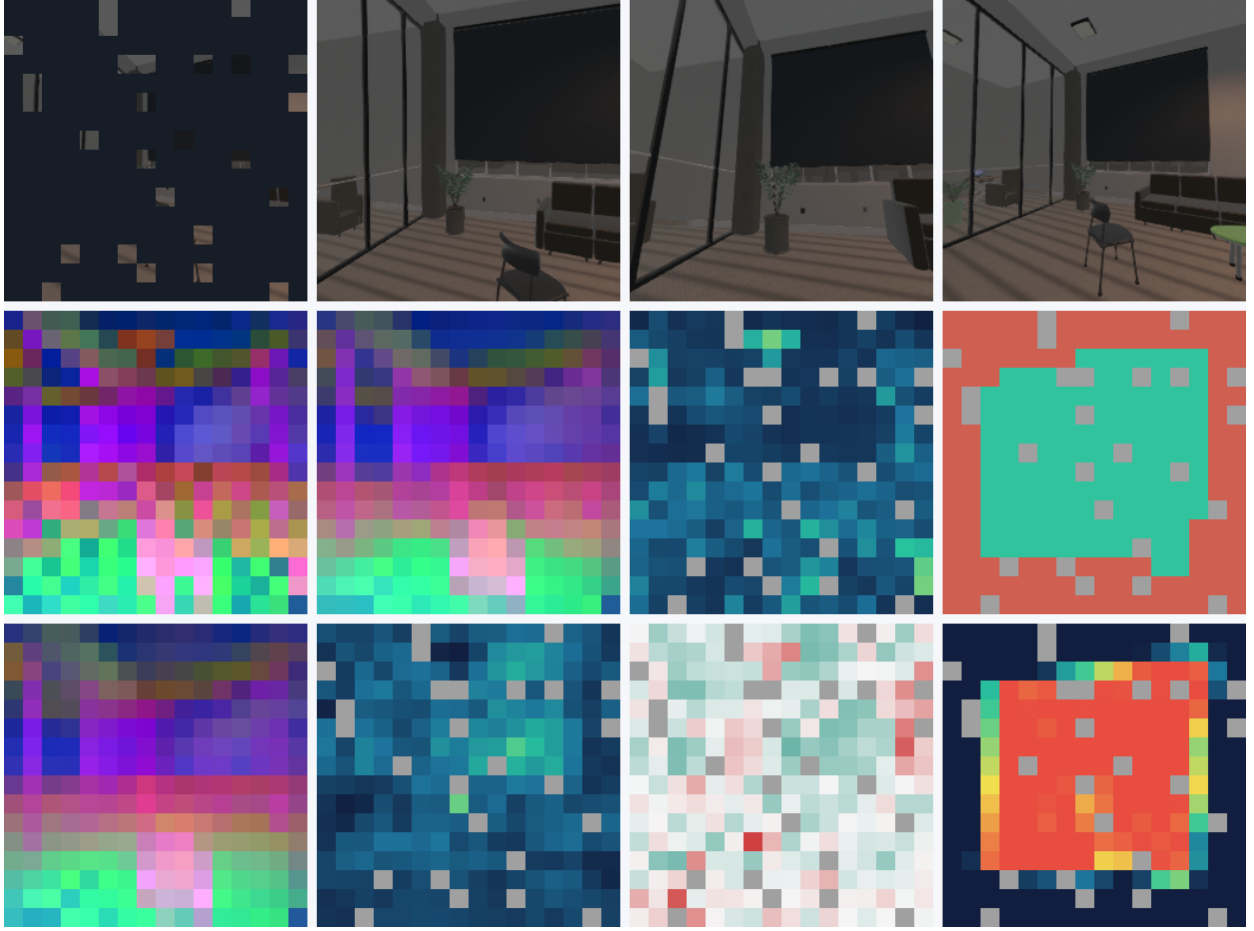
6.3 Room 2610091053, target view 0



Top row, left to right: sparse target input; reference 1; reference 2 (or an explicitly blank panel); full target RGB for evaluation only. Second row: teacher latent; predicted latent; hidden-token error; geometric co-visibility truth. If present, the third row shows the same model's monocular latent, learned RI score, signed reference benefit and geometric visibility fraction. RI/fraction maps use a 0–1 scale. Reference benefit is (monocular MSE minus cross-view MSE) divided by monocular MSE: -1 red (worse), 0 white, +1 teal (better), saturating outside that range. Unlike the clipped training RI target, this display retains negative effects. RI is an unconstrained regression score, not a calibrated visibility probability. The teacher and full target do not enter sparse inference.

Feature signal/error: 7.07 dB (not RGB PSNR). Hidden-token MSE: 0.196250. Cosine: 0.895862. Values are recomputed in Rust from the exported arrays and checked against the population evaluation. All samples use the same error and projection conventions.

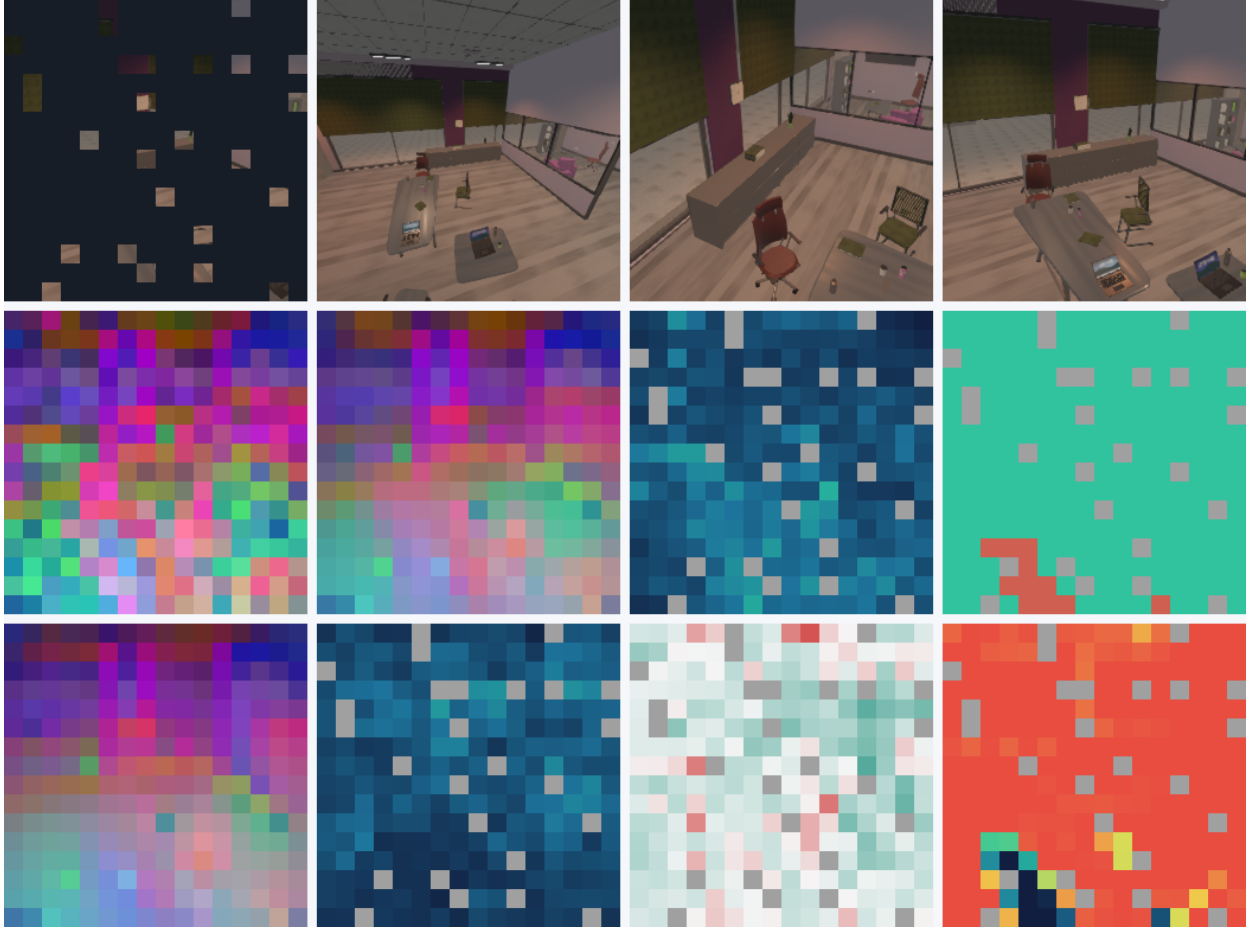
6.4 Room 2610091078, target view 2



Top row, left to right: sparse target input; reference 1; reference 2 (or an explicitly blank panel); full target RGB for evaluation only. Second row: teacher latent; predicted latent; hidden-token error; geometric co-visibility truth. If present, the third row shows the same model's monocular latent, learned RI score, signed reference benefit and geometric visibility fraction. RI/fraction maps use a 0–1 scale. Reference benefit is (monocular MSE minus cross-view MSE) divided by monocular MSE: -1 red (worse), 0 white, +1 teal (better), saturating outside that range. Unlike the clipped training RI target, this display retains negative effects. RI is an unconstrained regression score, not a calibrated visibility probability. The teacher and full target do not enter sparse inference.

Feature signal/error: 7.91 dB (not RGB PSNR). Hidden-token MSE: 0.161992. Cosine: 0.914849. Values are recomputed in Rust from the exported arrays and checked against the population evaluation. All samples use the same error and projection conventions.

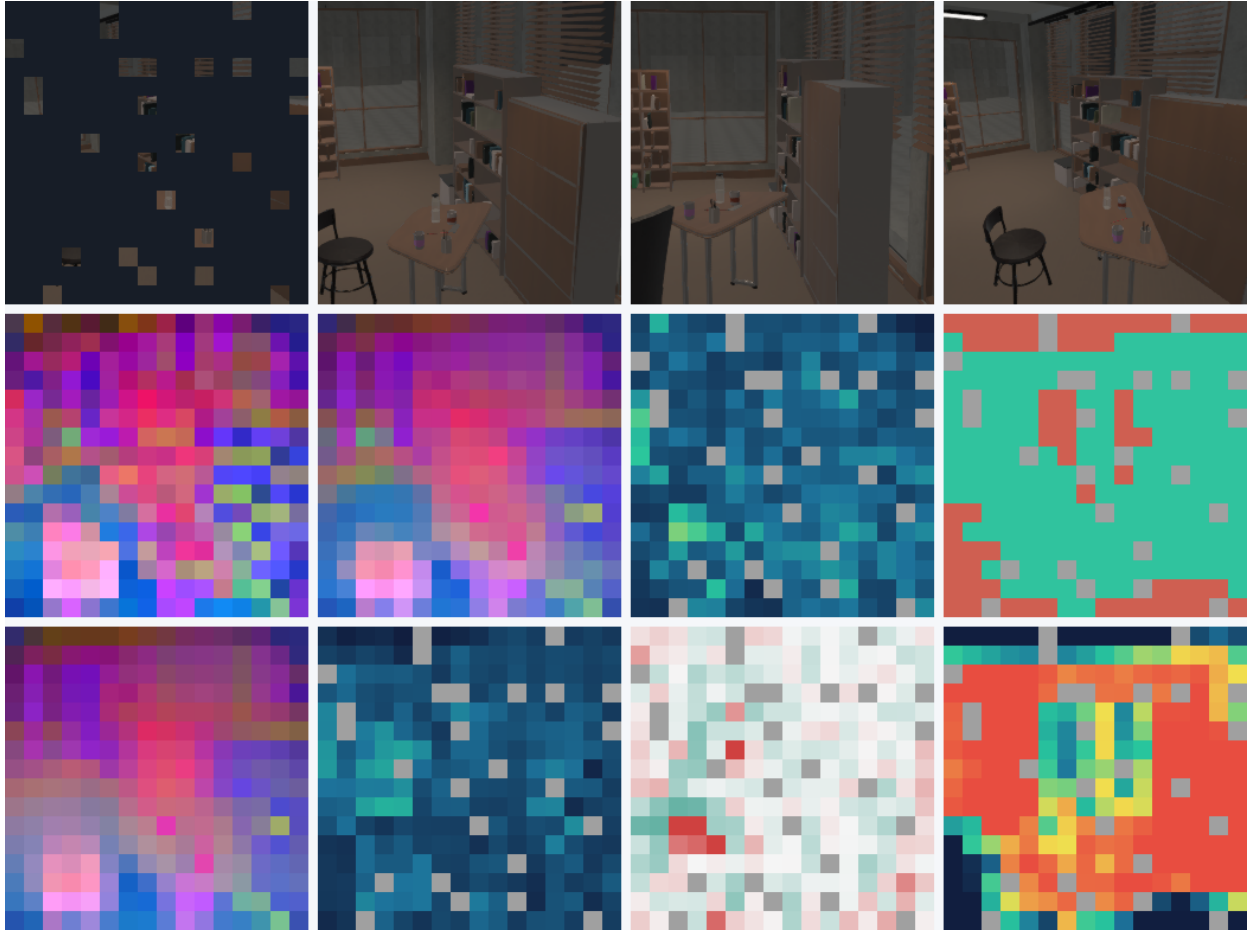
6.5 Room 2610091104, target view 0



Top row, left to right: sparse target input; reference 1; reference 2 (or an explicitly blank panel); full target RGB for evaluation only. Second row: teacher latent; predicted latent; hidden-token error; geometric co-visibility truth. If present, the third row shows the same model's monocular latent, learned RI score, signed reference benefit and geometric visibility fraction. RI/fraction maps use a 0–1 scale. Reference benefit is (monocular MSE minus cross-view MSE) divided by monocular MSE: -1 red (worse), 0 white, +1 teal (better), saturating outside that range. Unlike the clipped training RI target, this display retains negative effects. RI is an unconstrained regression score, not a calibrated visibility probability. The teacher and full target do not enter sparse inference.

Feature signal/error: 7.67 dB (not RGB PSNR). Hidden-token MSE: 0.171015. Cosine: 0.909975. Values are recomputed in Rust from the exported arrays and checked against the population evaluation. All samples use the same error and projection conventions.

6.6 Room 2610091129, target view 3



Top row, left to right: sparse target input; reference 1; reference 2 (or an explicitly blank panel); full target RGB for evaluation only. Second row: teacher latent; predicted latent; hidden-token error; geometric co-visibility truth. If present, the third row shows the same model's monocular latent, learned RI score, signed reference benefit and geometric visibility fraction. RI/fraction maps use a 0–1 scale. Reference benefit is (monocular MSE minus cross-view MSE) divided by monocular MSE: -1 red (worse), 0 white, +1 teal (better), saturating outside that range. Unlike the clipped training RI target, this display retains negative effects. RI is an unconstrained regression score, not a calibrated visibility probability. The teacher and full target do not enter sparse inference.

Feature signal/error: 6.92 dB (not RGB PSNR). Hidden-token MSE: 0.203265. Cosine: 0.892416. Values are recomputed in Rust from the exported arrays and checked against the population evaluation. All samples use the same error and projection conventions.

7 Training efficiency

Command duration 10.1 minutes; observed board energy 57.17 Wh; observed board joules per logged target 33.50; telemetry coverage 99.8%. Device-wide board power/utilization, including shared desktop load; observed intervals only, gaps over 5 seconds excluded. Process memory is separately attributed. Stage timing excludes first 10 updates when at least 21 exist; energy retains command preparation and evaluation. Utilization is sample-median, not SM occupancy.

Encoder stage	Median (ms)	p95 (ms)	Targets/s
2	1255.37	1302.47	12.75

Mean observed board power: 340.68 W. These are warmed per-update rates; complete-command energy retains preparation, validation and checkpoint overhead. Peak training-process VRAM: 49110 MiB. Peak

training-process RSS: 7285 MiB.

Encoder adaptation. Verified training stages: 384 updates with full encoder trainable (steps 1-384). Logged encoder gradient counts are checked against each stage. These counts describe optimization, not evidence of improved transfer.

Encoder feature preservation. Final encoder features are anchored to a pinned frozen ancestor with loss weight 4.0. Training feature RMS drift is 6.86% in the first and 8.44% in the last 32-update window (100 times the square root of mean per-image, energy-normalized feature MSE, equally weighted over sparse/full target and reference routes). Intermediate geometric features have no direct preservation target. This is a training regularization diagnostic, not RGB error or independent accuracy. The frozen anchor is absent from inference.

Renderer-supervised correspondence. Renderer-supervised correspondence is a training-only auxiliary (weight 0.1, temperature 0.07). First versus last 32-update mean NLL: 2.768 to 2.419 nats; usable query fraction: 46.5% to 47.1%. These are changing-minibatch diagnostics. Geometry and camera labels are absent from RGB inference.

Training convergence diagnostic. First versus last 32-update mean correspondence NLL: pair conditioning 1.5758 to 1.5008; same-image conditioning 1.6442 to 1.5560. These are descriptive minibatch statistics, not an independent generalization estimate. Known-transform descriptor supervision is motivated by SiLK [6]; implementation is independent, and no SiLK code or weights are used.

8 Limitations and qualification requirements

- This publication binds one frozen foundation checkpoint and one final attached-head training run. Head weight hashes, optimizer records, native validation predictions and training logs are pinned separately. It does not compare separately trained private model versions.
- The 800-update head phase uses 32 training rooms and 8 disjoint validation rooms from existing synthetic data, with one seed. These are development cohorts previously used by the foundation study, not a fresh generalization test. The final checkpoint is fixed by the schedule, not chosen on validation performance.
- Numerical head stability passed, but camera generalization is poor: training camera regression loss 0.000131 versus validation 0.387406; validation rotation 11.64 degrees, translation direction 32.68 degrees and focal relative error 44.35%. The calibration head is not deployment-ready.
- The camera head uses dense target/reference RGB pair features. RGB completion uses only sparse target and reference information. A centered principal point is assumed; translation scale is not predicted. No camera labels enter either inference path.
- RGB PSNR uses hidden sRGB pixels in [0,1] and averages per-target PSNR. It is separate from feature signal/error dB. The same RGB decoder scores the cross-view and monocular routes. PSNR gains do not establish sharp details, geometric consistency or real-scene completion quality.
- The latent gallery and external correspondence/solver results describe the unchanged foundation checkpoint. Attached-head metrics use a separate explicitly labeled 24-target cohort. Legacy calibrated RANSAC results use known intrinsics and are not measurements of the learned calibration head.
- The training curve and board-efficiency panel describe the historical foundation phase. The attached heads train on CPU in 66.44 seconds total; frozen GPU feature export takes 44.99 seconds under the old remaining allowance. Desktop GPU activity is included in board observations. No process-attributed power claim.
- A legacy assessment provenance flag incorrectly said `geometry_training_supervision=false`. Foundation training used renderer-supervised correspondence. The preserved training configuration and label-cache records are authoritative; the later exporter correction passed exact replay.
- Encoder/fusion joint training with these output heads has not been qualified. Camera intrinsics generalization, additional seeds, larger datasets, strict numerical reference-order invariance, independent real data and matched public baselines remain open. SOTA and blur resolution are not established.
- No noncommercial pretrained weights or teachers enter this lineage. Depth and other future heads remain untrained. Earlier recorded failed gates and numerical/exporter limitations are not waived by this head study.

9 Reproducibility and extension

The model library, trainer, data reader, evaluation and publication are separate Rust crates. A new decoder/head supplies a checkpoint-bound capability record with explicit metric units, population, aggregation and evaluation status. Camera scoring includes SO(3) angular error, signed translation-direction error, normalized focal error and pose AUC at 5, 10 and 20 degrees. Zero ground-truth baselines are excluded from translation-direction and pose denominators; a zero predicted translation at a valid baseline is a failure. A trained head requires the separate checkpoint-bound optimization and prediction evidence reported in its capability section.

Generate this page and report from one TOML experiment manifest with `gekko-report build`. The bundle carries input and output hashes, exact sample identities, a common latent projection and the resolved report data. Local preparation does not commit, push, deploy or upload the repository. Checkpoints from old code identities require their sealed binaries for exact optimizer resume; weights-only new phases must record explicit ancestry and optimizer resets.

10 Related work and provenance

The project studies sparse visual encoding and multi-view fusion motivated by V-JEPA 2.1 and Gekko. Procedural room captures use published bevy_zeroverse packages. Calibrated 3D positional encodings remain a research extension. The optional supervised calibration head uses two continuous rotation columns, following the representation of Zhou et al. [8]; it does not implement or claim the published camera systems referenced below. The external matching protocols use the documented HPatches and ETH3D assets associated with ZeroCo [7], with the local input resolution and readout explicitly declared. This is not reproduction of its complete published matching system.

References

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- [5] ZipSplat: Fewer Gaussians, Better Splats. <https://arxiv.org/abs/2606.05102>.
- [6] P. Gleize, W. Wang and M. Feiszli. SiLK: Simple Learned Keypoints. <https://arxiv.org/abs/2304.06194>.
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- [8] Y. Zhou et al. On the Continuity of Rotation Representations in Neural Networks. CVPR 2019. <https://arxiv.org/abs/1812.07035>.
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